EG-CMFNet: An Edge-Guided Cross-Modal Fusion Network for Remote Sensing Semantic Segmentation

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***Abstract*—Multimodal remote sensing data often exhibit complementary information about ground objects and are widely utilized for semantic segmentation tasks. However, spatial heterogeneity and misalignment from different modalities hinder the utilization of multimodal information. To address these challenges, an edge-guided cross-modal fusion network (EG-CMFNet) is proposed for multimodal remote sensing imagery. We introduce a multimodal adaptive feature interaction (MAFI) module to adaptively interact with modal information through dual-stage spatial–channel interactions. Furthermore, during the feature fusion stage, a lightweight feature fusion (LFF) module suppresses unimodal noise to achieve efficient fusion of multimodal complementary information. Additionally, an edge auxiliary (EA) module is designed to extract boundary features precisely, mitigating confusion between semantic and edge information. Comparative experiments demonstrate that EG-CMFNet achieves state-of-the-art accuracy and robustness across three datasets of varying scales, including WHU-OPT-SAR, Dongying, and Pohang datasets. Specifically, EG-CMFNet achieves the highest mean intersection over union of 56.05%, 94.68%, and 97.51%, as well as overall accuracy (OA) of 84.47%, 94.97%, and 97.72%, respectively. The source code is available at https://github.com/masurq/EG-CMFNet**

***Index Terms*—CNN, edge information, multimodal remote sensing data, semantic segmentation, Transformer.**

I. Introduction

REMOTE sensing image semantic segmentation enables efficient large-scale Earth observation (EO) by performing pixel-level classification of the imagery, which is widely applied in fields such as land-cover classification [1], [2], urban planning [3], and environmental monitoring [4]. With the development of satellite remote sensing technology for EO, there is an increase in remote sensing data, such as optical, synthetic aperture radar (SAR), and digital surface model (DSM), providing richer feature representations for land cover.

As an important data source for remote sensing images, optical images are the earliest applied for land-cover interpretation due to their rich spectral information. It distinguishes land cover types by utilizing spectral reflectance characteristics and spatial textural features. Optical imagery is less sensitive to the structural information of ground objects, and its effectiveness is limited under adverse weather conditions. SAR captures both intensity and phase information of ground objects due to its imaging mechanism, enabling the extraction of structural details [5], [6]. Moreover, it operates independently of weather conditions and illumination. Compared with single-modal data, multimodal data can provide complementary information from different sensing modalities. For example, optical imagery is well-suited for capturing spectral details, while SAR is more effective at extracting structural features. The combination of both modalities facilitates more accurate land-cover segmentation [7].

Research on multimodal remote sensing imagery segmentation is primarily divided into traditional methods and deep learning approaches. Traditional methods use physical models to extract features from multimodal images and implement classification via machine learning classifiers to obtain the classification map [8], [9]. While offering strong interpretability, they often exhibit limited feature extraction capabilities and generalization ability when handling heterogeneous data with varying resolutions and scales [10]. Thanks to continually advancing satellite sensors, the abundant remote sensing data provide a solid foundation for deep learning applications. This has spurred the shift of multimodal semantic segmentation methods from traditional machine learning to deep learning approaches for large-scale data processing. Early deep learning methods primarily used a single-stream architecture, integrating multimodal data through early fusion [11], [12]. However, these methods rely on strict cross-modal feature alignment and frequently suffer from modality imbalance. In contrast, multistream architectures utilize separate branches to process heterogeneous modal data, thereby enabling more effective processing of information from different modalities [13], [14].

Based on different model architectures, deep learning approaches are mainly categorized into CNN- and

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Transformer-based methods. CNN-based methods efficiently extract multiscale features through the local receptive fields of convolutional kernels and downsampling, enabling efficient automatic learning of feature representations. However, CNNs are inherently limited to local receptive fields by convolution operations and predominantly focus on local information [15], [16]. Some studies expand receptive fields via large-kernel convolutions and multiscale integration strategies [17], [18]; however, these approaches still suffer from high computational overhead and loss of local details. With advancements in deep learning, Transformer-based methods have been increasingly adopted for feature extraction in remote sensing imagery [19], [20]. Their ability to capture global dependencies and model long-range contextual relationships makes them well-suited for extracting global image features. Nevertheless, the global computation mechanism of Transformers often overlooks fine-grained local details. Consequently, the integration of complementary strengths from CNNs and Transformers, which leverages both local and global information, enables effective land-cover segmentation.

Despite significant advances in multimodal semantic segmentation, three critical issues remain, as outlined next.

1) *Limitations of Modal Heterogeneity and Spatial Misalignment:* The heterogeneity between optical and SAR data introduces distinct nonlinear radiometric properties, which constrain the extraction of advantageous features inherent to individual modalities [21]. To mitigate heterogeneity in ground object representation across modalities, some studies propose correcting bimodal features before fusion [22], typically adopting a dual-stage processing strategy: first dressing single-modal information in spatial dimension and then aggregating cross-modal features via interaction mechanisms. However, this design neglects the issue of spatial misalignment between modalities, compromising the intrinsic channel-wise and spatial interrelationships. Thus, research on mechanisms addressing intermodal heterogeneity remains underexplored.
2) *Representation Discrepancy and Limitations of Computational Inefficiency in Cross-Modal Fusion:* Inconsistent spatial scales and feature representations of remote sensing data impede effective information aggregation through direct fusion. To address multimodal fusion challenges, two different primary approaches are explored. One category involves element-wise summation or channel-wise concatenation, as exemplified by Resunet-a [23], which directly combines multimodal images at the input stage via concatenation to form a single tensor. This primitive fusion approach fails to exploit cross-modal semantic relationships effectively and tends to cause feature confusion or information loss. The other category comprises attention-based approaches. Models such as SA-Gate [24] and CMX [25] leverage cross-attention for modality weight calculation, enabling effective selective fusion with suppressed intermodal noise. Although cross-attention mechanisms effectively capture interdependencies across modalities, they incur high computational complexity. Furthermore, when confronted with varying scales and morphologies of ground objects, the global operations of cross-attention mechanisms may compromise fine-grained local details.
3) *Limitations of Insufficient Utilization of Edge Information:* Most current segmentation methods primarily focus on extracting high-level semantic features (e.g., long-range dependencies and global context). However, they often overlook the role of edge information, resulting in blurred object boundaries [26]. Increasingly complex internal structures, enhanced intraclass heterogeneity, and intensified interclass confusion in remote sensing images highlight the need for accurate edge features to achieve precise semantic segmentation. Current studies typically enhance segmentation accuracy by fusing local edge features with global semantic information to achieve edge-guided feature refinement [27], [28]. However, reliance on semantic branch features for edge extraction causes semantic-edge confusion, degrading segmentation performance.

Based on the above analyses, to address fusion challenges and edge blurring issues, we propose a novel edge-guided cross-modal fusion network (EG-CMFNet). Specifically, a multistream CNN and Transformer architecture extracts global semantic features from optical and SAR data. Meanwhile, multimodal adaptive feature interaction (MAFI) modules and lightweight feature fusion (LFF) modules conduct adaptive cross-modal interactions and efficient multiscale fusion across hierarchical feature levels. Additionally, a CNN-based edge branch incorporates an edge auxiliary (EA) module to refine the boundary by leveraging edge information. Our contributions are summarized as follows.

1) EG-CMFNet is a novel multimodal semantic segmentation framework that employs a CNN and Transformer hybrid architecture. This architecture enables the extraction of both edge details and global semantic features while effectively integrating complementary information across multiple modalities. Extensive experiments conducted on three datasets of varying scales validate the effectiveness of the proposed method for multimodal semantic segmentation tasks.
2) To address feature misalignment in multimodal data, we propose an MAFI module. It employs a spatial–channel dual-stage adaptive interaction mechanism to reduce intermodal discrepancies, with adaptive parameters modulating interactions between modalities.
3) We propose an LFF module that fuses multimodal information through noise reduction and a multilevel fusion mechanism. Compared with other fusion mechanisms, LFF achieves a balance between accuracy and computational complexity.
4) We propose an edge guidance mechanism integrated into the multimodal segmentation framework. A dedicated branch processes edge information separately to prevent interference between global semantic features and edge features. Meanwhile, an EA module captures multiscale edge features in multiple directions to mitigate boundary blurring.

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The structure of this article is as follows. Section II reviews the related works in multimodal semantic segmentation. Section III presents the structure of the proposed EG-CMFNet, with a full description of each component. Section IV provides extensive experiments and thorough analyses. Section V discusses the proposed method in detail. Finally, Section VI provides a summary of our work.

II. Related Work

A. CNN-Based Multimodal Semantic Segmentation Methods

CNNs effectively achieve consistent feature extraction across modalities owing to their efficient spatial feature extraction capability and the local connectivity of convolutional kernels. They have been widely applied in multimodal semantic segmentation. PSTNet [29] utilizes segmentation confidence maps from optical images as auxiliary information. Concatenating these maps with optical and thermal infrared images reinforces discriminative features in raw data. However, this makes it difficult to leverage distinctive information across modalities. FuseNet [12] adopts two CNN branches to extract features from optical and depth data separately, incorporating partial depth network features into the optical branch. However, this approach tends to cause confusion between the information from the two branches. AFNet [30] extracts optical and thermal infrared image features at the deepest encoder layer and fuses them via an attention fusion module to enhance discrimination of confusing ground objects. While these deep interactive features establish intermodal connections, the utilization of cross-modal global information remains limited. CMGFNet [31] employs two separate encoders to generate multilevel features for optical and DSM modalities. It achieves high-precision building extraction through cross-modal and multilevel feature fusion. SA-Gate [24] proposes a separate and aggregate operation to alternately merge features from both modalities for recalibration. Thereby, it selectively aggregates information from the two modalities for segmentation. While the aforementioned methods yield satisfactory performance in multimodal semantic segmentation, they are hampered by the limited receptive fields of convolutional kernels. This curtails their capability to capture semantic features beyond local pixel neighborhoods, further impairing their capacity to harvest global contextual information and model long-range dependencies.

B. Transformer-Based Multimodal Semantic Segmentation Methods

Transformer [32] transcends the limitations of local receptive fields by employing a multihead self-attention mechanism to extract global contextual information. This capability has significant advantages in extracting global semantic information for multimodal semantic segmentation, which in turn has accelerated its widespread adoption in recent research. CMFNet [33] leverages the capability of the Transformer to learn long-range dependencies in fused features, enhancing the capture of discriminative representations from optical and DSM data. It also reduces decoding ambiguity to some extent. CMX [25] adopts a ViT-based architecture to achieve cross-modal feature interaction and fusion. Unified cross-modal interactions enable different branches to effectively leverage complementary information from adjacent modalities. FTransUNet [34] enhances interfeature dependency learning through a cross-scale fusion module that integrates multilevel multimodal features. However, this approach lacks a feedback mechanism between branches to enable cross-modal interaction. SiMultiF [35] adopts a dual-branch Siamese structure to extract features. Adaptive feature weighting deeply extracts multimodal data, thereby optimizing cross-modal interaction. Other methods, including MCANet [36] and TMFNet [37], primarily utilize attention modules in the encoder stage to calculate the weights of different modal features. Although these methods selectively adjust cross-modal features via weight computation, they often overlook feature independence and spatial discrepancies between optical and SAR data, leading to similar feature weighting representations across both modalities.

C. Edge-Guided Semantic Segmentation Methods

The above methods primarily focus on extracting global semantic features but show limited capability in capturing edge features. These designs often overlook the significance of edge details, resulting in degraded segmentation performance at object boundaries and poor local detail preservation. Researchers have proposed numerous approaches to address these issues in edge-based segmentation. Traditionally, edge detection methods have been classified into three main categories. The first category includes first-order gradient operators, such as the Sobel operator [38], which leverage the magnitude of the image's first-order gray-level derivative to localize edges. These methods offer computational efficiency but are sensitive to noise. The second category comprises second-order derivative operators, like the Laplacian operator [39], which detect edges via zero-crossings in the second-order derivative of gray-level intensity. Combining Gaussian smoothing with the Laplacian convolution enhances detection in regions of rapid intensity change and suppresses noise, albeit at increased computational complexity. The third category consists of adaptive threshold operators, such as the Canny operator [40], which dynamically adjust edge detection based on local statistical characteristics. This approach yields better edge continuity but exhibits cumbersome parameter tuning. Nevertheless, these traditional edge detection operators exhibit limited representational capacity, rely on fixed designs, and lack adaptive adjustment capabilities. In contrast, researchers increasingly integrate deep learning models with edge information to produce more precise edge segmentation results. Chen et al. [41] propose an edge-aware convolution kernel that leverages geometric information within channels to more effectively extract multimodal image features. Gao et al. [42] integrate semantic segmentation with optimizing boundary supervision to extend multimodal semantic segmentation scenarios. Sui et al. [43] propose a novel BIBED-Seg framework that incorporates predicted object boundaries as prior inputs to the semantic segmentation network, yielding boundary-optimized segmentation outcomes. Zhou et al. [44] propose an edge-aware guidance fusion

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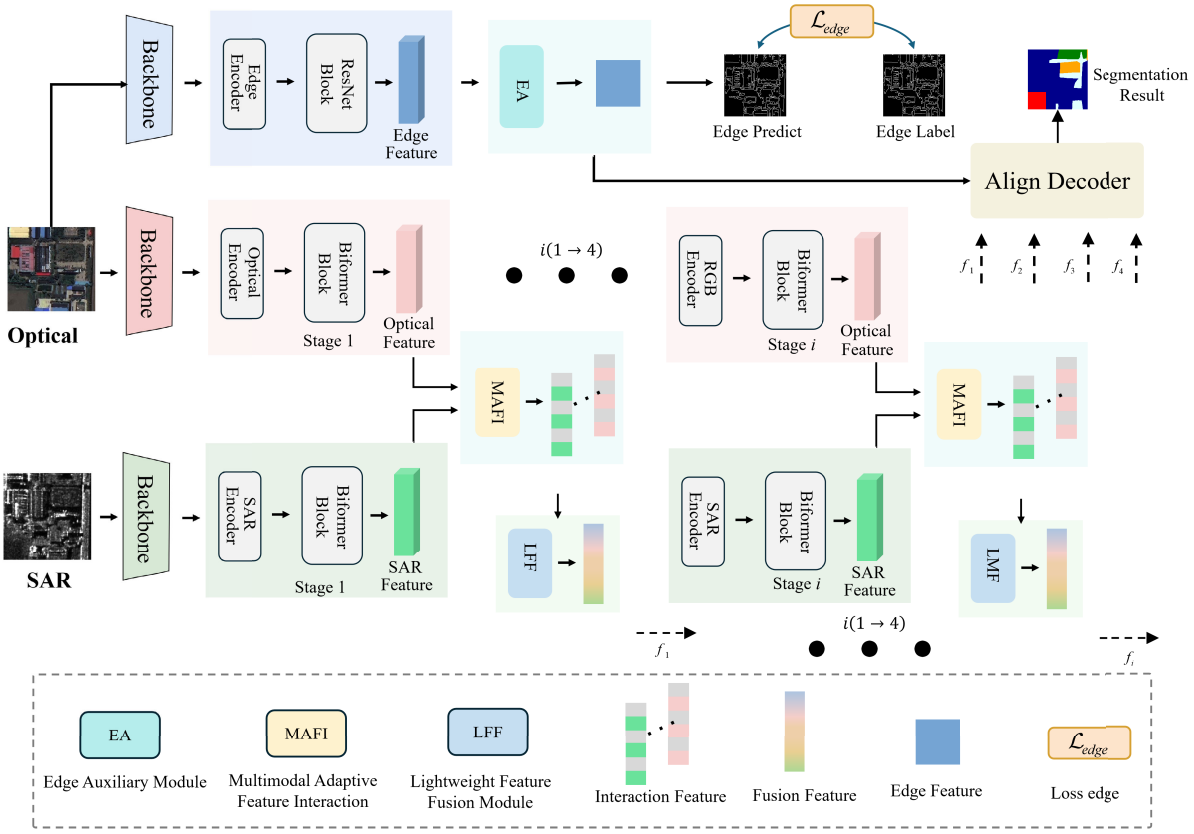


Fig. 1. Architecture of the proposed EG-CMFNet. The framework consists of five main components: 1) an EA module that extracts edge information; 2) a multimodal encoder that extracts features from optical and SAR images; 3) an MAFI module that facilitates cross-modal information exchange; 4) an LFF module that integrates multimodal features; and 5) a decoder that combines the fused features and edge information to produce the final classification results.

network that leverages boundary information to refine multimodal feature interactions. On this basis, Dong et al. [45] introduce EGFNet, which employs ConvNeXt for feature extraction. Experiment results demonstrate that more effective boundary extraction can improve cross-modal segmentation performance. Current methods typically leverage edge predictions as prior inputs for semantic networks or employ specialized loss functions to enhance segmentation accuracy. However, they often overlook intraclass heterogeneity between object boundaries and interiors, leading to confusion between semantic and edge features that limits the effective use of edge information.

In this study, we propose an edge-guided cross-modal fusion framework designed to enhance segmentation performance. The framework leverages complementary multimodal information and edge features through carefully designed feature interaction, feature fusion, and an edge branch.

III. Proposed Method

This section provides a comprehensive overview of the EG-CMFNet, then introduces its key modules in detail, and finally presents the hybrid loss function.

A. Overall Architecture

The proposed EG-CMFNet employs an encoder–decoder architecture comprising five components: a multistream encoder, an MAFI module, an LFF module, an EA module, and a decoder, as shown in Fig. 1. Specifically, the multistream encoder utilizes a pseudo-Siamese network to extract multilevel features from optical and SAR data separately. Both the optical and SAR branches adopt Biformer [46] as a backbone for the feature extraction. Although the two backbones share identical network architectures, they do not share weights due to different imaging mechanisms and data distribution characteristics of optical and SAR images. To effectively capture modality-specific characteristics and facilitate deep feature interaction across modalities, the MAFI module is introduced between adjacent stages of the backbone network. The interacted features are subsequently fused by the LFF module to generate cross-modal representations. The edge branch adopts ResNet-18 as its backbone. Unlike the Biformer architecture, ResNet-18 focuses on local textures that are crucial for boundary extraction. The EA module adaptively refines boundary features extracted from the CNN encoder and performs multiscale fusion to enhance edge detail and boundary precision. Finally, the fused semantic features and edge information are jointly input to the multistage decoder, which progressively upsamples and aggregates features to generate the final segmentation result.

B. MAFI Module

Due to the distinct imaging mechanisms of optical and SAR sensors, the acquired data often exhibit significantly

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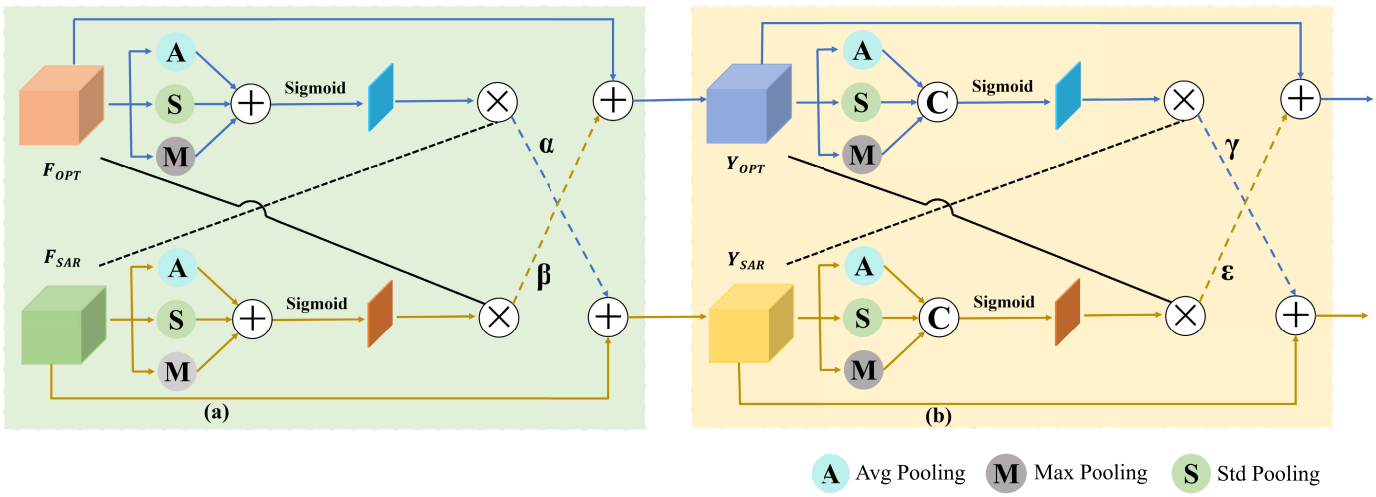


Fig. 2. Architecture of the MAFI module. It implements cross-modal interaction through (a) channel interaction and (b) spatial interaction, thereby enhancing the correlation between spectral features and geometric positions across modalities.

different characteristics [47]. SAR images contain phase and intensity information, demonstrating unique advantages in characterizing ground object structures. In contrast, optical images provide rich spectral and texture information, which is particularly valuable for remote sensing image interpretation [48], [49]. It is precisely this divergence in physical principles that leads to heterogeneous feature representations for identical or similar land-cover categories across the two modalities. Additionally, geometric imaging discrepancies between optical and SAR sensors can induce spatial misalignment where target objects exhibit pixel-level displacements due to divergent sensor perspectives [50]. Conventional feature interaction methods often overlook the complementarity of multimodal information in aligning spatial features and may disrupt the inherent correlations between spatial and channel dimensions. Such inherent modality heterogeneity and spatial shifts compromise the representational capacity of individual modal features, thereby posing significant challenges to subsequent fusion processes.

To enhance the correlation of feature representations and spatial positions for identical land classes across modalities, we propose an MAFI module to address the limitation that channel-wise interaction alone fails to correct intramodal feature misalignments, which in turn hinder cross-modal correlation between identical land-cover classes. The MAFI module performs spatial recalibration of features through cross-modal interactions, thereby achieving pixel-wise alignment between SAR and optical features. The detailed architecture is shown in Fig. 2, with the channel interaction stage procedures described as follows.

1) Channel-Wise Interaction Stage: As illustrated in Fig. 2(a), the feature maps of optical and SAR are denoted as $\mathbf{F}_{OPT} \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{F}_{SAR} \in \mathbb{R}^{C\times H\times W}$. Here, C, H, and W denote the number of channels, height, and width of the feature maps, respectively. The subscripts represent the corresponding modality. A spatial comprehensive pooling strategy is applied to both modalities' features. It integrates global average pooling to capture holistic characteristics, global standard deviation pooling to quantify feature distribution dispersion, and global max pooling to highlight locally salient features. This strategy captures richer salient feature information through multiple pooling. Subsequently, the pooled global feature vectors are fed into a multilayer perceptron (MLP) for nonlinear transformation to derive more abstract feature representations. The resulting global feature vectors are then summed and processed through a sigmoid activation function to generate channel attention maps $\mathbf{S}_{OPT}$ and $\mathbf{S}_{SAR}$, which incorporate global contextual information. As shown in the following equation:

$$\begin{cases} \mathbf{S}_{SAR} = \sigma\left(\mathrm{MLP}\left(\mathrm{Avg}\left(\mathbf{F}_{SAR}\right)\right) + \mathrm{MLP}\left(\mathrm{Std}\left(\mathbf{F}_{SAR}\right)\right) \right. \\ \qquad\qquad \left. + \mathrm{MLP}\left(\mathrm{Max}\left(\mathbf{F}_{SAR}\right)\right)\right) \\ \mathbf{S}_{OPT} = \sigma\left(\mathrm{MLP}\left(\mathrm{Avg}\left(\mathbf{F}_{OPT}\right)\right) + \mathrm{MLP}\left(\mathrm{Std}\left(\mathbf{F}_{OPT}\right)\right) \right. \\ \qquad\qquad \left. + \mathrm{MLP}\left(\mathrm{Max}\left(\mathbf{F}_{OPT}\right)\right)\right) \end{cases} \tag{1}$$

where Avg(·), Max(·), and Std(·) correspond to global average pooling, global max pooling, and global standard deviation pooling, respectively, and $\sigma(\cdot)$ denotes the sigmoid activation function

$$\mathrm{MLP} = \mathrm{Linear}\left(C_{\mathrm{mid}}, C_{\mathrm{out}}\right)\left(\mathrm{ReLU}\left(\mathrm{Linear}\left(C_{\mathrm{in}}, C_{\mathrm{mid}}\right)(\cdot)\right)\right). \tag{2}$$

The MLP defined in (2) consists of two linear layers and a ReLU activation function. Linear($C_{\mathrm{in}}, C_{\mathrm{mid}}$) denotes the linear layers, C_{in} and C_{mid} represent the input and output channel dimensions, and ReLU(·) denotes the ReLU activation.

To enable cross-modal feature interaction, the attention maps $\mathbf{S}_{OPT}$ and $\mathbf{S}_{SAR}$, representing optical and SAR information, respectively, are element-wise multiplied with the original SAR and optical features. This is an attention-guided gating mechanism rather than a direct multiplication of raw optical and SAR features. The learned cross-modal attention maps selectively suppress modality-specific noise while preserving consistent semantic features. This gating mechanism effectively enhances discriminative representations and reduces inconsistency across modalities. This operation leverages complementary modality information to refine the original feature

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representations, thereby enhancing the correlation between optical and SAR features. Adaptive calibration parameters are introduced to suppress modality-inherent discrepancies. Residual connections are incorporated to prevent gradient explosion, generating the channel-interacted feature maps $\mathbf{Y}_{OPT}$ and $\mathbf{Y}_{SAR}$ as formalized in the following equation:

$$\begin{cases}\mathbf{Y}_{OPT} = \beta \cdot \mathbf{S}_{SAR} \odot \mathbf{F}_{OPT} + \mathbf{F}_{OPT} \\ \mathbf{Y}_{SAR} = \alpha \cdot \mathbf{S}_{OPT} \odot \mathbf{F}_{SAR} + \mathbf{F}_{SAR}\end{cases} \tag{3}$$

where α and β are learnable parameters that dynamically modulate the fusion weights, enhancing the representation of discriminative features while suppressing modality-specific noise. This mechanism prioritizes informative feature representations, mitigates irrelevant cross-modal interference, and ultimately achieves optimal fusion performance.

2) Spatial-Wise Interaction Stage: Following the initial cross-modal interaction at the channel-wise stage, spatial interaction is introduced to rectify spatial shifts between modalities, achieving feature alignment across modalities. Instead of enforcing explicit geometric transformations, this stage performs implicit feature alignment by learning correspondences through attention-driven modulation at the downsampled feature scale. In this way, regions with cross-modal inconsistency are assigned lower weights to suppress their influence, while structurally consistent areas are assigned higher weights to promote cross-modal alignment. As illustrated in Fig. 2(b), the channel-interacted feature maps $\mathbf{Y}_{OPT}$ and $\mathbf{Y}_{SAR}$ undergo comprehensive channel-wise pooling. This process retains channel information from the preceding stage while simultaneously extracting spatially salient features across modalities. The resulting global feature vectors are then concatenated and aggregated via a convolutional layer and generate global attention maps $\mathbf{S}^S_{SAR}$ and $\mathbf{S}^S_{OPT}$ via a sigmoid activation function. These maps integrate spatial, positional, and channel characteristics, as formulated in the following equation:

$$\begin{cases}\mathbf{S}^S_{SAR} = \sigma\left(f^{7\times7}\left(\text{Concat}\left(\text{Avg}\left(\mathbf{Y}_{SAR}\right), \right.\right.\right. \\ \qquad \left.\left.\left.\text{Std}\left(\mathbf{Y}_{SAR}\right), \text{Max}\left(\mathbf{Y}_{SAR}\right)\right)\right)\right) \\ \mathbf{S}^S_{OPT} = \sigma\left(f^{7\times7}\left(\text{Concat}\left(\text{Avg}\left(\mathbf{Y}_{OPT}\right), \right.\right.\right. \\ \qquad \left.\left.\left.\text{Std}\left(\mathbf{Y}_{OPT}\right), \text{Max}\left(\mathbf{Y}_{OPT}\right)\right)\right)\right)\end{cases} \tag{4}$$

where $f^{7\times7}(\cdot)$ denotes a 7×7 convolution.

In the spatial-wise interaction phase, a second cross-modal interaction is conducted. We adopt the same operations to integrate multimodal information and introduce new learnable parameters to further mitigate intermodal noise interference, yielding spatially interacted feature maps $\mathbf{X}_{OPT}$ and $\mathbf{X}_{SAR}$. The mathematical formulation is shown in the following equation:

$$\begin{cases}\mathbf{X}_{OPT} = \varepsilon \cdot \mathbf{S}^S_{SAR} \odot \mathbf{Y}_{OPT} + \mathbf{Y}_{OPT} \\ \mathbf{X}_{SAR} = \gamma \cdot \mathbf{S}^S_{OPT} \odot \mathbf{Y}_{SAR} + \mathbf{Y}_{SAR}\end{cases} \tag{5}$$

where ε and γ are learnable parameters.

C. LFF Module

The complementary characteristics of SAR and optical data provide multidimensional information. Fusing the complementary features significantly improves segmentation performance. Existing multimodal feature fusion methods mainly include two types: summation or concatenation-based approaches [12], [51], [52] and cross-attention mechanisms [24]. The former adopts simple concatenation or feature map summation for fusion but may exacerbate modality-specific noise. In contrast, cross-attention mechanisms effectively model intermodal correlations, but their intrinsic function of managing cross-modal interactions incurs higher computational overhead and structural complexity [53], [54]. In particular, the quadratic computational complexity of cross-attention hinders its compatibility with high-resolution remote sensing images, where large variations in object scale can lead to loss of fine-grained details [55], [56]. To balance effectiveness and computational efficiency in multimodal feature fusion, we propose an LFF module tailored to the characteristics of high-resolution remote sensing imagery. This module comprises three components: a raw image enhancement path, a multiscale feature extraction path, and a feature fusion path, with its detailed architecture shown in Fig. 3.

Feature fusion across modalities often introduces integrated noise, such as salt-and-pepper noise in optical images and speckle noise in SAR data. To mitigate such integrated noise, we introduce the SimAM attention mechanism [57] into the raw image enhancement path. SimAM evaluates the importance of each neuron by minimizing an energy function derived from linear separability theory. Informative neurons typically exhibit activation patterns that are distinguishable from their surroundings, while uninformative neurons are suppressed. SimAM serves as a denoising filter that downweights unstable noisy regions, ensuring that subsequent convolutional fusion operates on more reliable and discriminative features. The energy function used for neuron evaluation is defined in the following equation:

$$\begin{cases}e^*_t = \dfrac{4\left(\hat{\sigma}^2 + \lambda\right)}{(t-\hat{\mu})^2 + 2\hat{\sigma}^2 + 2\lambda} \\ \hat{\mu} = \dfrac{1}{M}\displaystyle\sum_{i=1}^{M} x_i \\ \hat{\sigma}^2 = \dfrac{1}{M}\displaystyle\sum_{i=1}^{M}(x_i - \hat{\mu})^2\end{cases} \tag{6}$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ denote the mean and variance of all neurons excluding the target neuron t. A lower energy value indicates higher distinctiveness and importance of the neuron t relative to its surrounding neurons. Based on the minimum energy obtained from (6), the importance of a neuron can be determined by $\frac{1}{E}$. After evaluating neuron importance, the feature matrix is refined according to the following equation:

$$\tilde{X} = \text{sigmoid}\left(\frac{1}{E}\right) \times X \tag{7}$$

where E aggregates e^*_t across both channel and spatial dimensions.

Compared with traditional image fusion methods, cross-attention mechanisms can effectively capture global contextual information but often induce the degradation of local texture and edge details [58], [59]. In contrast, the local perception characteristics of convolutional structures better preserve

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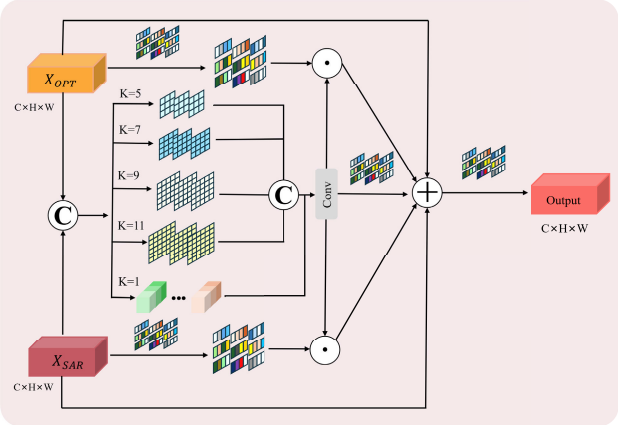


Fig. 3. Architecture of the LFF module. It employs SimAM attention to enhance salient features across both modalities, utilizes multiscale depthwise separable convolutions to enable cross-modal interaction and capture global context, and finally fuses the aggregated features to produce the output.

object boundary integrity. To further enhance the adaptability of convolutional feature extraction to complex remote sensing scenes, particularly when facing large variations in object scale and shape, we design a multiscale feature extraction path to capture structural and textural information at different resolutions. Specifically, we process concatenated SAR and optical image features using multiscale depthwise convolution (DWConv) to simultaneously capture global semantics and local details, subsequently reorganizing features via 1×1 pointwise convolution within a multiscale parallel computing architecture. The feature maps of optical and SAR images after the feature interaction module are denoted as $\mathbf{X}_{OPT} \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{X}_{SAR} \in \mathbb{R}^{C\times H\times W}$. The input multimodal feature maps are first concatenated. Four DWConv with varying kernel sizes are employed to target change regions of different scales. This captures intramodal contextual relationships, ultimately yielding multiscale feature maps X_c^k where k = 5, 7, 9, 11. Subsequently, the four convolutional outputs are concatenated along the channel dimension. Weighted combination and feature map reconstruction are performed via 1×1 pointwise convolution integrated with standard convolution, ultimately yielding the target feature map $\mathbf{X}_f$ as formulated in the following equations:

$$\mathbf{X}_c^k = \mathrm{DWConv}_{k\times k}\left(\mathrm{Concat}\left(\mathbf{X}_{OPT}, \mathbf{X}_{SAR}\right)\right) \quad k = \{5, 7, 9, 11\} \tag{8}$$

$$\mathbf{X}_f = \mathrm{CBR}_{1\times 1}\left(\mathrm{Conv}_{1\times 1}^{pw}\left(\mathrm{Concat}\left(\mathbf{X}_c^5, \mathbf{X}_c^7, \mathbf{X}_c^9, \mathbf{X}_c^{11}\right)\right)\right) \tag{9}$$

$$\mathrm{CBR} = \mathrm{ReLu}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times 1}\right)\right) \tag{10}$$

where k denotes the convolution kernel size, $\mathrm{Conv}_{1\times 1}^{pw}$ represents the pointwise convolution, and subscripts of $\mathrm{CBR}_{1\times 1}$ indicate the dimensions of convolutional kernels employed.

Although concatenated features encompass global characteristics of intermodal relationships, modality-specific discriminative features are also beneficial to complement them. To enhance discriminative responses and suppress redundant information, we use SimAM to generate modality-specific weight maps from the original optical and SAR features. In parallel, we construct a global representation $\mathbf{X}_f$ by concatenating multiscale features from the two modalities. We then reweight $\mathbf{X}_f$ with the modality-specific weight maps through element-wise multiplication, which yields $\widetilde{x_{opt}}$ and $\widetilde{x_{sar}}$. The formulation is given as follows:

$$\begin{cases} \widetilde{x_{opt}} = \mathrm{SimAM}\left(\mathbf{X}_{OPT}\right) \odot \mathbf{X}_f \\ \widetilde{x_{sar}} = \mathrm{SimAM}\left(\mathbf{X}_{SAR}\right) \odot \mathbf{X}_f. \end{cases} \tag{11}$$

The aggregated features are subsequently processed by SimAM to generate the final result, achieving a simple yet effective fusion of multimodal features as formulated in the following equation:

$$\mathbf{X}_{\text{fusion}} = \mathrm{SimAM}\left(\mathrm{CBR}_{3\times 3}\left(\mathbf{X}_{OPT} + \mathbf{X}_{SAR} + \widetilde{x_{opt}} + \widetilde{x_{sar}} + \mathbf{X}_f\right)\right). \tag{12}$$

Moreover, beyond the degradation of local details caused by global operations, the high computational complexity of cross-attention mechanisms poses challenges for their integration into lightweight fusion frameworks. To address this issue, we adopt a structurally simpler approach to reduce computational overhead. As specified in (13), given input feature maps $\mathbf{F}_{OPT} \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{F}_{SAR} \in \mathbb{R}^{C\times H\times W}$, the computational cost of cross-attention comprises three components: Q/K/V projection operations, attention-weighted summation, and output linear transformation, yielding an overall complexity of $O(N^2)$. In contrast, the computational cost of the reconstructed fusion mechanism proposed in this article, formulated in (14), consists of three parts: the raw input path computation, multiscale feature extraction path computation, and feature fusion computation, with the total complexity optimized to $O(N)$

$$\mathrm{FLOP}_{\text{attn}} = 2 * \left(3NC^2 + 2N^2C + NC^2\right) \tag{13}$$

$$\mathrm{FLOP}_{\text{LFF}} = 8C^2N + 180CN + 4C^2N \tag{14}$$

where $N = H \times W$ denotes the total number of pixels and C represents the number of feature channels.

D. EA Module

Semantic segmentation and boundary detection are intrinsically linked tasks [60]. Pixels within a ground object exhibit homogeneity, contrasting sharply with those at boundaries [61]. In multimodal remote sensing semantic segmentation scenarios, ambiguous object boundaries and intraclass heterogeneity directly result in irregular object contours and omission of small-scale targets. Concurrently, the misalignment of cross-modal features between optical and SAR images further exacerbates boundary localization uncertainty [62]. These shortcomings adversely affect the accuracy of the segmentation results. Current methodologies predominantly emphasize the internal consistency of ground objects, often insufficiently accounting for the intricate details of their boundaries. The potential significance of boundary information in the semantic segmentation of remote sensing images is frequently overlooked. Therefore, we introduce an edge branch that extracts more discriminative edge information to provide stable boundary guidance for semantic segmentation.

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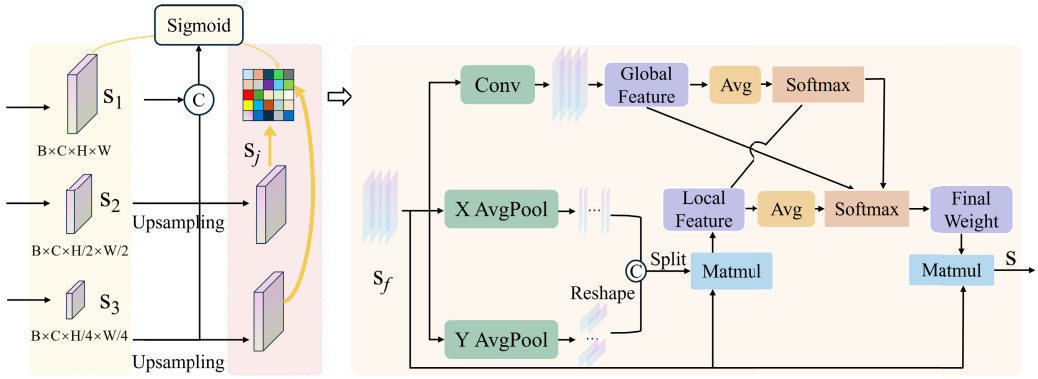


Fig. 4. Architecture of the EA module. Features from the first three layers of ResNet (S_1–S_3) are concatenated; multiscale feature fusion is performed via an attention mechanism and fed to subsequent layers that integrate local and global features, ultimately yielding edge detail information.

This branch is designed to improve boundary prediction with limited computational cost. Although SAR imagery contains rich structural information, speckle noise makes these boundaries discontinuous. By contrast, boundary textures in optical images are usually clearer and more continuous [63]. Accordingly, by balancing boundary reliability and computational efficiency, we adopt CNN backbones such as ResNet to extract optical boundary cues, thereby improving boundary localization accuracy and further refining the segmentation results. As illustrated in Fig. 4, since low-level features contain richer detail [64], we upsample and concatenate the first three feature maps (S_1–S_3) to capture multiscale edge information of ground objects. Furthermore, EMA [65] is proposed to integrate local edge features with global semantic features. This integration enables a distinct separation between ground object boundaries and complex backgrounds and produces well-defined boundary contours. Specific implementation details are described as follows.

Let $\mathbf{S}_i \in \mathbb{R}^{C\times\frac{H}{i}\times\frac{W}{i}}$ $\{i = 1, 2, 3\}$ represent the features extracted from the first to three layers of the ResNet. These multiscale features are upsampled to a common spatial resolution and concatenated to obtain a feature map $\mathbf{S}_c^u \in \mathbb{R}^{C\times H\times W}$. To mitigate confusion caused by multiscale edge detail variations, a weighted attention mechanism is used to adjust feature fusion across scales, yielding the final aggregated multiscale result $\mathbf{S}_f$. The specific equations are as follows:

$$\mathbf{S}_c^u = \text{Concat}\left(\mathbf{S}_1, \text{Upsample}\left(H\times W\right)\left(\mathbf{S}_2, \mathbf{S}_3\right)\right) \tag{15}$$

$$\mathbf{S}_j = \text{Softmax}\left(\text{Conv}_{1\times1}\left(\mathbf{S}_c^u\right)\right) \tag{16}$$

$$\mathbf{S}_j^i = \text{Split}\left(\mathbf{S}_j\right) \tag{17}$$

$$\mathbf{S}_f = \sum_{i=1}^{3} \mathbf{S}_i \times \mathbf{S}_j^i \tag{18}$$

where Upsample($H \times W$) denotes the feature map upsampled to size H × W and Split(·) represents splitting feature maps along the channel dimension for uniform segmentation.

To address the problem of edge blurring in remote sensing images caused by the complex boundaries and noise associated with small objects, we propose a method that integrates local edge features with global context. This is accomplished by capturing information both along the x- and y-axes, as well as from a global perspective, to combine local features with broader context. The multiscale aggregated result $\mathbf{S}_f$ is grouped to form feature maps $\mathbf{S}_g \in \mathbb{R}^{B\times g\times\frac{C}{g}\times H\times W}$, where B denotes the batch size and g the group count. Subsequent 1-D horizontal and vertical global pooling operations along the x- and y-axes, respectively, produce concatenated outputs. Convolution and channel grouping operations then remap local information from both directions into the refined feature maps $\mathbf{S}_h'$ and $\mathbf{S}_w'$, as expressed in the following equations:

$$\begin{cases} \mathbf{S}_g = \text{Group}\left(\mathbf{S}_f\right) \\ \mathbf{S}_h = \text{Avg_h}\left(\mathbf{S}_g\right) \\ \mathbf{S}_w = \text{Avg_w}\left(\mathbf{S}_g\right) \end{cases} \tag{19}$$

$$\mathbf{S}_{hw} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{S}_h, \mathbf{S}_w\right)\right) \tag{20}$$

$$\mathbf{S}_h', \mathbf{S}_w' = \text{Split}\left(\mathbf{S}_{hw}\right) \tag{21}$$

where Group(·) represents the feature grouping operation and Avg_h(·) and Avg_w(·) denote horizontal and vertical global pooling, respectively.

The integration of local detail features with global semantic features for boundary contour generation is achieved using an EMA attention mechanism. This mechanism enables interaction across different positions while preserving spatial information. Specifically, the local feature map is processed by a sigmoid function to generate attention weights, which are then multiplied with the original grouped features to yield the locally aggregated feature $\mathbf{S}_{local}$. Meanwhile, the original grouped features are reconstructed through convolution into the global feature $\mathbf{S}_{global}$. The locally aggregated and global features are then element-wise multiplied and summed with attention weights produced by a Softmax function to obtain the fused weights. Finally, these weights are multiplied with the original grouped features, integrating local and global information to derive the edge feature $\mathbf{S}$. The specific equations are as follows:

$$\mathbf{S}_{local} = \text{Gn}\left(\text{Sigmoid}\left(\mathbf{S}_h'\right) \odot \text{Sigmoid}\left(\mathbf{S}_w'\right) \odot \mathbf{S}_g\right) \tag{22}$$

$$\mathbf{S}_{global} = \text{Conv}_{3\times3}\left(\mathbf{S}_g\right) \tag{23}$$

$$\begin{aligned} \mathbf{S} = \mathbf{S}_g \times \Big(&\left(\text{Softmax}\left(\text{Avg}\left(\mathbf{S}_{local}\right)\right) \odot \mathbf{S}_{global}\right) \\ &+ \left(\text{Softmax}\left(\text{Avg}\left(\mathbf{S}_{global}\right)\right) \odot \mathbf{S}_{local}\right)\Big) \end{aligned} \tag{24}$$

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where Gn(·) denotes group normalization and ⊙ denotes the Hadamard product.

The edge feature map **S** is upsampled to half the size of the original input image to derive the final boundary feature, while a boundary prediction map is generated via the Softmax function. The boundary feature is then fed into the final decoder layer and fused with semantic branch features to produce the prediction map.

E. Loss Function

The loss function employed in this study comprises a semantic loss and an edge loss. Specifically, the semantic loss combines a cross-entropy loss and a Dice loss. The former is suitable for multiclass segmentation tasks, while the latter optimizes for regional consistency and mitigates class imbalance [66]. In addition, since the joint learning of semantic segmentation and edge detection demands accurate boundary delineation, we also adopt the Dice loss as the edge loss to directly maximize the overlap between the predicted and ground-truth edge maps. The semantic loss L_{seg} and edge loss L_{edge} are formulated as follows:

$$L_{\text{seg}} = L_{\text{ce}} + L_{\text{dice}} \tag{25}$$

$$L_{\text{edge}} = L_{\text{dice}} \tag{26}$$

where L_{ce} denotes the cross-entropy loss and L_{dice} denotes the Dice loss, which are defined as follows:

$$L_{\text{ce}} = -\sum_{ij}\sum_{k=1}^{K} y_{ij,k} \log p_{ij,k} \tag{27}$$

$$L_{\text{dice}} = 1 - \frac{2\,|\hat{y} \cap y|}{|\hat{y}| + |y|} \tag{28}$$

where $\hat{y}$ denotes the network predicted result, y represents the ground truth, and K indicates the number of classes. Subscripts i and j denote the pixel at row i and column j. Subscript k denotes the class index.

The total loss L_{total} is expressed as follows:

$$L_{\text{total}} = \frac{L_{\text{edge}}\left(F_{ep}, F_{el}\right) + L_{\text{seg}}\left(F_{sp}, F_{sl}\right)}{2} \tag{29}$$

where F_{sp} and F_{ep} represent the segmentation prediction and edge prediction and F_{sl} and F_{el} represent the segmentation label and edge label.

IV. Experiments and Analysis

To evaluate the performance of the proposed EG-CMFNet for multimodal remote sensing image semantic segmentation, we conduct experiments on three public datasets: WHU-OPT-SAR [36], Pohang [67], and Dongying [67]. Comprehensive ablation studies are also carried out to analyze the contribution of each module in the proposed EG-CMFNet.

A. Datasets

They are given as follows.

1) *WHU-OPT-SAR:* The WHU-OPT-SAR dataset, a publicly accessible multisource remote sensing resource provided by Wuhan University, comprises 100 optical–SAR image pairs covering approximately 50 000 km^2 in Hubei, China. Each image measures 5556 × 3704 pixels at a 5-m spatial resolution, presenting diverse terrain features with seven annotated classes: farmland, forest, urban, village, water, road, and others. Following the original dataset's splitting strategy, we use 80 images for training and 20 for testing. During preprocessing, images are cropped into nonoverlapping 512 × 512 subimages.
2) *Pohang:* The Pohang dataset, presented by Ren et al. [67], is a publicly accessible multimodal remote sensing dataset containing 7831 optical and SAR image pairs. The experimental data are acquired from the Pohang region in South Korea. Optical images are acquired from the GF-2 satellite, while SAR data are derived from the GF-3 satellite and resampled to 1-m spatial resolution. All images are cropped to 256 × 256 pixels. The dataset comprises five land-cover classes: buildings, road, greenery, water, and farmland. For model training, validation, and testing, we randomly divide the dataset into training, validation, and test subsets at a 5:3:2 ratio.
3) *Dongying:* The Dongying dataset, as another multimodal resource from Ren et al. [67], is the smallest among the three datasets. Both modalities are resampled to 1-m spatial resolution and cropped to 256 × 256 pixels. This dataset includes the same land-cover classes as the Pohang dataset. For model training, validation, and testing, we randomly divide the dataset into training, validation, and test subsets at a 5:3:2 ratio.

B. Implementation Details

In this study, all experiments are conducted on an NVIDIA RTX 4090 GPU using PyTorch. To accelerate convergence, we employ the AdamW optimizer [68] with a weight decay of 0.02. The learning rate follows a combined WarmUp and MultiStepLR schedule, initializing at 5e^{-5}. After the WarmUp, the rate progressively increases to 1e^{-4} at fixed epochs (five for WHU-OPT-SAR and ten for the Pohang and Dongying datasets). The main data augmentation methods include random scaling, cropping, and flipping. Given divergent dataset scales, parameters differed; WHU-OPT-SAR uses 60 epochs with a batch size of 6. In contrast, Pohang and Dongying employ 100 epochs at a batch size of 16.

C. Comparative Models

To comprehensively validate the performance of the proposed EG-CMFNet, we compare it with several of the latest semantic segmentation methods.

1) *V_FuseNet (2018) [51]:* A CNN-based multimodal fusion method that integrates LiDAR and multispectral data through feature addition and compares early and late fusion strategies through comparative analysis.
2) *SA-Gate (2020) [24]:* A CNN-based multimodal feature fusion network that filters noise and recalibrates RGB-D features via a separation-aggregation gating mechanism, demonstrating superior performance in RGB-D semantic segmentation tasks.

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3) *AFNet (2020) [30]:* A multipath CNN encoder extracts multisource features and addresses the challenge of fusing multipath and multilevel features through feature concatenation and attention mechanisms.
4) *CFNet (2022) [69]:* A cross-modal CNN for joint optical–SAR land-cover classification that designs four fusion modules and incorporates gating modules to enhance feature complementarity via bidirectional information flow.
5) *MCANet (2022) [36]:* This CNN framework employs a pseudo-Siamese convolutional network, a multimodal cross-attention module, and a multiscale feature fusion module for joint semantic segmentation of optical and SAR data. Additionally, it constructs the first large-scale optical–SAR dataset, WHU-OPT-SAR.
6) *CMGFNet (2022) [31]:* This network employs dual CNN encoders to extract multilevel features from RGB and DSM data. It achieves cross-modal fusion through gated fusion modules, integrating high- and low-level features via a top-down strategy.
7) *CMX (2023) [25]:* This dual-branch Transformer network employs cross-modal feature rectification and fusion modules to achieve efficient fusion of RGB, depth, or LiDAR data and address modality-specific constraints in conventional methods.
8) *GFBN (2024) [42]:* A Transformer-based multimodal fusion network employs cross-modal feature fusion and a boundary enhancement module to address challenges in feature fusion and detail preservation for multimodal data.
9) *CATNet (2024) [70]:* This Swin Transformer-based network extracts multiscale semantics and global context to enhance and fuse RGB-Depth features. It employs a cascaded correction decoder to suppress noise and compensate for information loss while incorporating an edge generator to refine boundary details.
10) *FtransUnet (2024) [34]:* This hybrid CNN–Transformer network employs a shallow fusion module to extract multiscale details. It utilizes a three-stage fusion Transformer to achieve cross-modal interaction and global modeling of deep semantic features, thereby addressing insufficient local and global information extraction in remote sensing multimodal fusion.

D. Evaluation Metrics

To ensure rigorous comparison with the aforementioned methods across three datasets, two evaluation metric categories are adopted. The first category assesses model accuracy, including overall accuracy (OA), mean intersection over union (mIoU), and $F1$-score [33]. The second category evaluates model complexity through computational metrics: floating point operations (FLOPs) and parameter count (Params). The formulas for computing OA, mIoU, and $F1$ are as follows:

$$\mathrm{OA} = \frac{\sum_{K=1}^{P} \mathrm{TP_K}}{\sum_{K=1}^{P} (\mathrm{TP_K} + \mathrm{FN_K})} \tag{30}$$

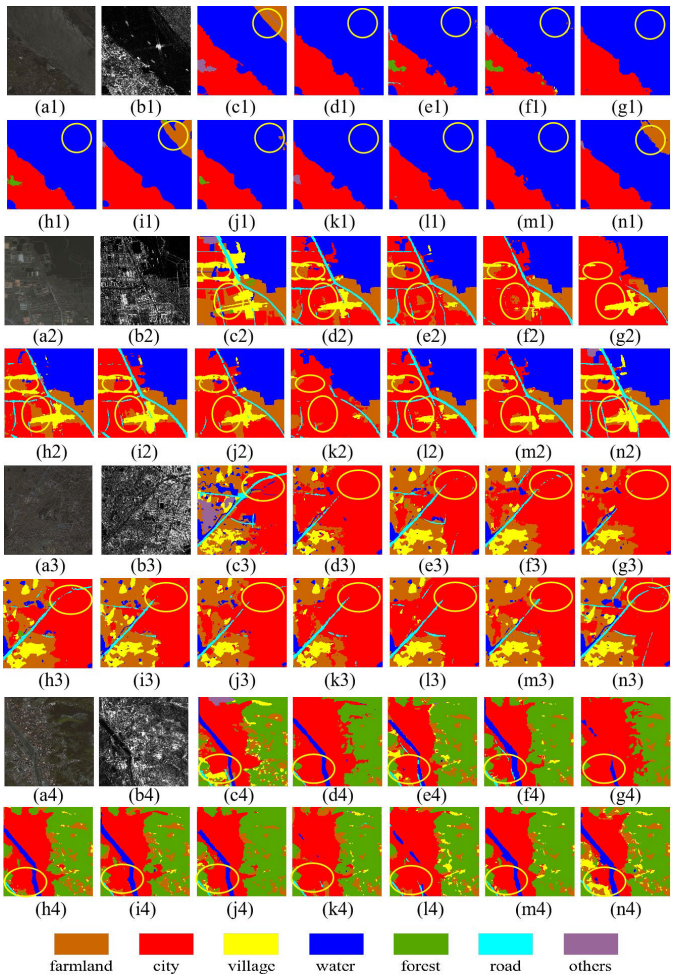


Fig. 5. Visualization of segmentation results on the WHU-OPT-SAR dataset. (a1)–(a4) Optical images. (b1)–(b4) SAR images. (c1)–(c4) Ground truth. (d1)–(d4) SA-Gate. (e1)–(e4) V_FuseNet. (f1)–(f4) CFNet. (g1)–(g4) MCANet. (h1)–(h4) CMGFNet. (i1)–(i4) CMX. (j1)–(j4) AFNet. (k1)–(k4) CATNet. (l1)–(l4) GFBN. (m1)–(m4) FtransUnet. (n1)–(n4) Proposed method.

$$\mathrm{mIoU} = \frac{1}{\mathrm{P}} \sum_{K=1}^{\mathrm{P}} \frac{\mathrm{TP_K}}{\mathrm{TP_K} + \mathrm{FP_K} + \mathrm{FN_K}} \tag{31}$$

$$\mathrm{F_1} = 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \tag{32}$$

where $\mathrm{TP_K}$ denotes true positive predictions, $\mathrm{TN_K}$ denotes true negative predictions, $\mathrm{FN_K}$ denotes false negative predictions, and $\mathrm{FP_K}$ denotes false positive predictions. Furthermore, precision and recall are given by the following formulas:

$$\mathrm{precision} = \frac{\mathrm{TP_K}}{\mathrm{TP_K} + \mathrm{FP_K}} \tag{33}$$

$$\mathrm{recall} = \frac{\mathrm{TP_K}}{\mathrm{TP_K} + \mathrm{FN_K}}. \tag{34}$$

E. Comparison Experiment

1) Performance Comparison on the WHU-Opt-Sar Dataset: Fig. 5 presents visual comparisons of all models on the WHU-OPT-SAR dataset, demonstrating significant advantages of the proposed approach in boundary preservation and handling mixed land-cover distributions. It can be observed from

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TABLE I
QUANTITATIVE COMPARISON RESULTS ON THE WHU-OPT-SAR DATASET

Method	F1-score (%)							OA (%)	mIoU (%)
	Farmland	City	Village	Water	Forest	Road	Others		
SA-Gate(2020)	79.7	71.4	64.9	73.2	90.4	54.6	22.2	81.92	51.41
V_FuseNet(2018)	79.4	70.4	60.2	73.7	90.7	51.3	28.8	82.35	50.82
CFNet(2022)	80.6	69.8	63.4	74.7	90.5	52.1	24.1	82.78	51.25
MCANet(2022)	81.1	67.6	64.2	77.4	90.7	52.1	22.4	83.17	51.52
CMGFNet(2022)	80.9	72.0	63.6	76.8	91.0	52.6	30.7	83.41	52.92
CMX(2023)	81.5	70.0	65.3	74.3	91.2	56.3	33.4	83.56	53.34
AFNet(2020)	81.1	71.5	64.1	77.2	91.0	54.1	31.4	83.61	53.29
CATNet(2024)	77.3	69.6	61.7	72.6	89.2	49.2	26.0	80.44	49.45
GFBN(2024)	79.8	69.7	63.9	74.9	90.7	53.7	30.2	82.40	52.02
FtransUnet(2024)	80.4	72.3	63.7	76.1	90.6	50.8	28.6	82.97	52.20
Proposed	**82.3**	**72.6**	**66.9**	**79.3**	**91.5**	**59.0**	**37.8**	**84.47**	**56.05**

the first and second rows of Fig. 5 that experimental region 1 primarily comprises city, water, and farmland classes due to sparse farmland distribution; most comparative models misjudge the top-right farmland region as water. Only CMX and EG-CMFNet accurately segment this farmland area. This capability may be primarily attributed to their incorporation of feature interaction modules before data fusion, thereby supplementing structural differences between farmland and water bodies in SAR data. Furthermore, EG-CMFNet employs a dual-stage adaptive interaction strategy. Through adaptive dynamic parameter adjustment, it suppresses oversensitivity to homogeneous optical textures while enhancing the discriminative expression of SAR features, significantly improving the accuracy of farmland segmentation. As shown in the third and fourth rows of Fig. 5, experimental region 2 exhibits complex land-cover characteristics dominated by city areas and water bodies with interspersed farmland and village complexes. The results reveal substantial misclassification and omission across comparative methods in complex environments, such as unsegmented villages and roads in yellow circle annotated regions, along with farmland misclassification. Compared to V_FuseNet and CFNet, which adopt simplistic sum-based fusion schemes neglecting intermodal interference, EG-CMFNet achieves superior segmentation performance for small-to-medium area targets, including farmland and villages. This capability originates from its lightweight multipath fusion module that employs CNN architectures to capture local features and fuses high backscatter from buildings in SAR data with optical textures and spectral features, effectively mitigating confusion among city, village, and farmland categories. Additionally, the model delivers exceptional performance in boundary integrity preservation for linear targets like roads. This advantage likely stems from the edge branch supplying constraints to the semantic branch, reducing mis-segmentation caused by ambiguous boundaries. Experimental region 3 is a city area comprising four land-cover classes: city, villages, roads, and farmland. As shown in the fourth and fifth rows of Fig. 5, existing methods exhibit fragmentation and misclassification issues in road segmentation within city areas. This stems from the limited resolution of remote sensing images, where roads are easily confused with city features.

While GFBN and CATnet achieve relatively superior performance in segmenting fragmented roads through the boundary enhancement module, they retain substantial misclassification overall. In contrast, EG-CMFNet processes edge information independently through its edge branch, preventing interference from the mixing of other semantic information in optical images with edge features. It enhances boundary details when semantic features are insufficient, significantly reducing land-cover boundary blurring and misclassification. Notably, manual labeling errors exist in roads at the lower left corner of experimental region 2 and parts of region 3. By leveraging edge feature guidance, EG-CMFNet enhances the differentiation between object interiors and boundaries, achieving precise road segmentation despite noisy labels and validating its robustness when faced with labeling errors. Experimental region 4 encompasses city river regions and sparsely distributed villages. As visualized in rows 7 and 8 of Fig. 5, EG-CMFNet delivers better segmentation performance for water bodies and small irregular villages than existing methods, particularly in boundary and target completeness. This likely stems from the Transformer backbone enhancing feature extraction from SAR and optical data, fully harnessing high backscatter characteristics of villages in SAR data and textural features in optical data. Besides, the CNN-based edge branch effectively supplements boundary information. Both components independently extract semantic and edge features to improve edge segmentation. The final segmentation results effectively validate the efficacy of the CNN–Transformer architecture.

Quantitative results are presented in Table I, where EG-CMFNet outperforms all comparative methods in all metrics. It achieves significant improvements of 2.71% in mIoU and 0.86% in OA over the suboptimal methods CMX and AFNet. Furthermore, EG-CMFNet surpasses the suboptimal model by 0.8%, 0.3%, 1.6%, 1.9%, 0.3%, 2.7%, and 4.4% in $F1$-score for respective land-cover categories. These results validate the effectiveness of the proposed method in leveraging optical and SAR data to improve land-cover segmentation accuracy.

2) Performance Comparison on the Dongying Dataset: The Dongying dataset comprises five land-cover classes with buildings dominating the sample distribution and exhibiting marked spatial regularity. We conduct comparison experiments on the Dongying dataset to validate the effectiveness of the proposed model across different scenarios. Experimental

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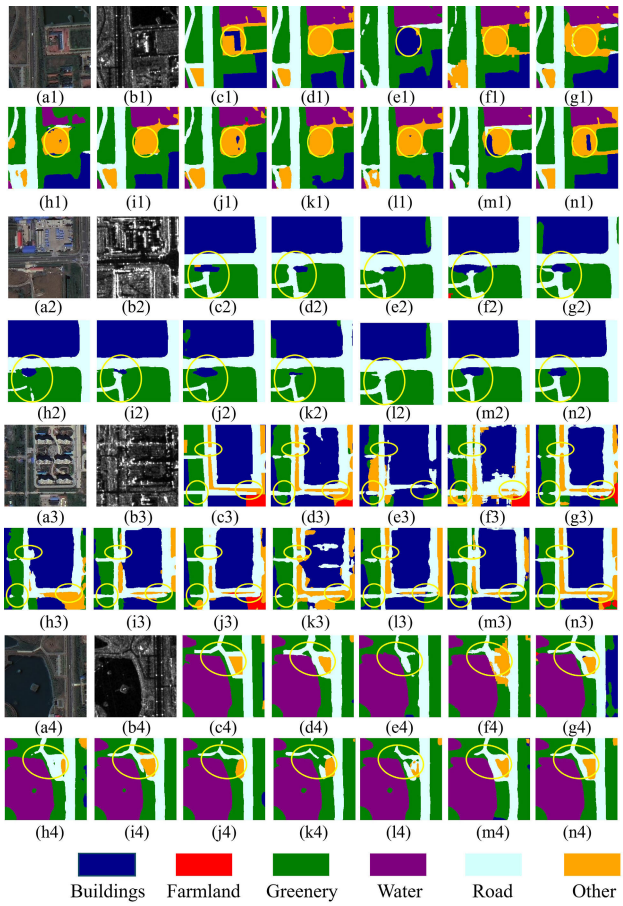


Fig. 6. Visualization of segmentation results on the Dongying dataset. (a1)–(a4) Optical images. (b1)–(b4) SAR images. (c1)–(c4) Ground truth. (d1)–(d4) SA-Gate. (e1)–(e4) V_FuseNet. (f1)–(f4) CFNet. (g1)–(g4) MCANet. (h1)–(h4) CMGFNet. (i1)–(i4) CMX. (j1)–(j4) AFNet. (k1)–(k4) CATNet. (l1)–(l4) GFBN. (m1)–(m4) FtransUnet. (n1)–(n4) Proposed method.

results of various models are shown in Fig. 6. To demonstrate the superiority of our method, we select four representative regions for detailed comparative analysis. Experimental region 1, located at the urban periphery, exhibits a complex feature distribution dominated by buildings, greenery, and water bodies. As shown in the first and second rows of Fig. 6, while CMX outperforms in segmenting building clusters, it demonstrates slight deficiencies in segmenting individual buildings. In contrast, EG-CMFNet achieves significantly superior performance in segmenting small-scale targets such as individual buildings. This enhancement stems from its novel feature interaction module. Unlike the channel-decoupled spatial aggregation mechanism adopted by CMX, this module fuses optical and SAR features across channel and spatial dimensions. By simultaneously leveraging textural characteristics from optical images and intensity information from SAR data, the model effectively segments small-scale targets, reducing omission and misclassification errors caused by the insufficiency of information from single-source data in complex backgrounds. Experimental region 2, an urban area with regular spatial patterns, primarily contains buildings, roads, and greenery. As illustrated in the third and fourth rows of Fig. 6, feature fusion-based methods (e.g., SA-Gate and MCANet) achieve superior overall segmentation performance, while edge enhancement methods such as GFBN and CATNet maintain finer boundary details. Compared to other methods, EG-CMFNet attains optimal segmentation performance. This is attributable to its multimodal fusion module fully exploiting complementary optical–SAR features, coupled with an edge augmentation module extracting multiscale edge characteristics. Consequently, the approach significantly enhances boundary delineation and structural integrity of segmented land-cover classes. As shown in rows 5 and 6 of Fig. 6, experimental region 3 features dense ground object distribution. Most methods exhibit interclass confusion between roads/greenery and farmland/greenery. MCANet and CMX utilize cross-attention-based feature fusion to mitigate confusion via global modeling, yet show inferior detailed segmentation. Conversely, EG-CMFNet introduces multiscale depthwise separable convolutions during feature fusion. This alleviates oversight in cross-attention mechanisms regarding detailed information while preserving intraclass consistency within complex scenes by leveraging discrepancies between global and local features.

Experimental region 4 is characterized by low spatial complexity and is dominated by water bodies, roads, and greenery. As shown in rows 7 and 8 of Fig. 6, V_FuseNet based on simple feature summation exhibits significant omission errors for the road class, and models employing basic attention mechanisms, CMGFNet and AFNet, show reduced omission but yield suboptimal segmentation performance. In contrast, methods based on feature interaction fusion, CMX, and SA-Gate, achieve superior results. This occurs because the linear geometric features of roads in optical images facilitate their distinction from areal features like water bodies, while their pronounced backscattering intensity in SAR data contrasts with other classes. Effective feature interaction proves crucial for achieving discriminative feature fusion. While CMX and SA-Gate enhance performance through fundamental interaction, the proposed method further optimizes feature rectification and interaction via a multilevel interaction mechanism, achieving optimal segmentation performance. This validates the superiority of deep interaction in excavating complementary information from multimodal data.

Quantitative results are presented in Table II. The proposed method outperforms comparative methods in both mIoU and OA, with improvements of 3.08% and 1.54%, respectively. For individual land-cover classes, it surpasses the second-best method (SA-Gate) by 1.0%, 0.8%, 1.9%, 1.2%, and 3.5% in $F1$-score, demonstrating generalization ability across diverse scenes with complex land-cover distributions.

3) Performance Comparison on the Pohang Dataset: To comprehensively evaluate model generalization, we conduct experiments on the Pohang dataset. By comparing the segmentation accuracy of EG-CMFNet against mainstream methods, we validate its performance in multisource remote sensing scenarios with relatively limited training samples. As shown in rows 1–4 of Fig. 7, comparative methods universally exhibit road edge fractures and segmentation discontinuities. Notably, GFBN maintains superior performance

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TABLE II
QUANTITATIVE COMPARISON RESULTS ON THE DONGYING DATASET

Method	F1-score (%)					mIoU (%)	OA (%)
	Buildings	Farmland	Greenery	Water	Road		
SA-Gate(2020)	97.2	97.9	94.7	96.7	91.4	91.6	95.97
V_FuseNet(2018)	93.4	93.3	84.7	93.4	85.1	82.02	90.26
CFNet(2022)	96.6	97.2	93.3	95.6	90.2	89.83	95.01
MCANet(2022)	96.7	97.6	93.3	96.3	91.1	90.61	95.47
CMGFNet(2022)	95.5	95.8	91.1	95.5	88.3	87.52	93.66
CMX(2023)	97.0	97.1	93.9	96.0	91.7	90.8	95.53
AFNet(2020)	97.1	97.5	94.3	96.5	91.4	91.19	95.78
CATNet(2024)	94.6	95.9	89.6	95.2	88.6	86.70	92.99
GFBN(2024)	95.1	95.3	89.7	95.1	88.0	86.47	93.07
FtransUnet(2024)	95.9	96.6	91.6	95.7	89.3	88.48	94.17
Proposed	**98.2**	**98.7**	**96.6**	**97.9**	**94.9**	**94.68**	**97.51**

TABLE III
QUANTITATIVE COMPARISON RESULTS ON THE POHANG DATASET

Method	F1-score (%)					mIoU (%)	OA (%)
	Buildings	Farmland	Greenery	Water	Road		
SA-Gate(2020)	99.6	98.2	89.9	97.1	97.6	93.40	97.10
V_FuseNet(2018)	99.3	95.7	79.7	91.4	94.1	85.99	93.56
CFNet(2022)	99.4	97.6	87.5	94.3	95.9	90.67	95.90
MCANet(2022)	99.5	98.1	88.9	96.6	97.3	92.69	96.84
CMGFNet(2022)	99.5	97.5	86.1	96.0	96.8	91.15	96.14
CMX(2023)	99.5	97.7	88.7	94.9	96.4	91.47	96.18
AFNet(2020)	99.5	97.6	86.7	95.6	94.1	91.07	96.01
CATNet(2024)	98.8	96.9	78.6	94.2	95.2	87.23	94.78
GFBN(2024)	99.4	96.5	84.3	93.0	94.7	88.36	94.61
FtransUnet(2024)	99.3	97.3	86.8	95.2	96.2	90.73	95.75
Proposed	**99.7**	**98.6**	**93.0**	**97.5**	**98.0**	**94.97**	**97.72**

across both regions due to its multisource feature fusion and boundary processing capabilities. However, in region 2, CATNet with analogous boundary modules underperforms MCANet. This stems from the region's relative simplicity, where efficient suppression of optical–SAR interference proves critical, while CATNet demonstrates weaker noise suppression than MCANet, specifically designed for optical–SAR data. In contrast, EG-CMFNet integrates triadic complementary strengths from optical images, SAR data, and edge information, significantly enhancing road geometric regularity and connectivity preservation while effectively mitigating segmentation continuity deficiencies. Experimental region 3 (rows 5 and 6 of Fig. 7) demonstrates that EG-CMFNet effectively suppresses oversegmentation of land cover compared to other methods, attributable to its joint feature interaction fusion mechanism, which enables the effective processing of multimodal semantic features while enhancing interclass separability. Segmentation results in experimental region 4 (rows 7 and 8 of Fig. 7) demonstrate prevalent undersegmentation in comparative methods, indicating their limited capability for multiscale objects. Our approach addresses this through multiscale global–local information fusion in both feature integration and boundary processing, achieving optimal segmentation outcomes.

Quantitative results in Table III demonstrate the superior segmentation accuracy of our method over comparative methods. Compared to the suboptimal SA-Gate, our network achieves gains of 1.57% in mIoU and 0.62% in OA. This robust performance across varying sample sizes validates the effectiveness of our method.

4) Module Ablation Experiments: To evaluate the effectiveness of each module in the EG-CMFNet, we conduct an ablation study in three datasets. Experimental results as shown in Table IV, where "✓" denotes utilization of the corresponding module. If the LFF is ablated, multimodal features are summed before decoder input. The results demonstrate that adding MAFI, LFF, or EA individually improves accuracy across all three datasets, validating the effectiveness of our proposed modules. For the single module, the EA module achieves the most significant improvement of 3.7% in mIoU over the baseline on the WHU-OPT-SAR dataset. On the other two datasets, the EA module also attains notable enhancements with improvements of 1.0% in mIoU, respectively, effectively demonstrating the importance of edge information. When incorporating dual modules, performance improvements on WHU-OPT-SAR are consistent, each yielding a 4.4% improvement in mIoU over the baseline. For the Dongying dataset, the integration of EA and MAFI achieves optimal efficacy with a 1.7% mIoU enhancement, attributable to its smaller scale and intricate terrain structures necessitating precise boundary segmentation; this synergy effectively reinforces interclass boundary localization while preserving structural integrity in complex features. Conversely, on the Pohang dataset, superior results emerged from MAFI-LFF or LFF-EA combinations, as its smallest scale and expansive land parcels prioritize intraclass distinction integrity, whereas multimodal complexities demand robust fusion mechanisms to harness complementary information, thus enabling LFF-based combinations to optimize segmentation outcomes. Furthermore, employing all modules collectively achieves optimal accuracy across all

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TABLE IV
ABLATION STUDY FOR THE MAFI/LFF/EA ON THE WHU-OPT-SAR/POHANG/DONGYING DATASETS

MAFI	LFF	EA	WHU-OPT-SAR		Dongying		Pohang	
			mIoU (%)	OA (%)	mIoU (%)	OA (%)	mIoU (%)	OA (%)
			51.2	82.2	92.1	96.2	92.6	96.8
✓			54.5	84.1	92.6	96.5	93.3	97.0
	✓		54.6	84.2	92.6	96.4	93.7	97.2
		✓	54.9	83.9	93.1	96.8	93.6	97.2
✓	✓		55.6	84.4	93.3	96.8	94.1	97.4
✓		✓	55.6	84.2	93.8	97.1	93.9	97.3
	✓	✓	55.6	84.2	93.6	97.0	94.0	97.4
✓	✓	✓	**56.0**	**84.5**	**94.7**	**97.5**	**95.0**	**97.7**

Note: ✓ denotes introducing the corresponding module.

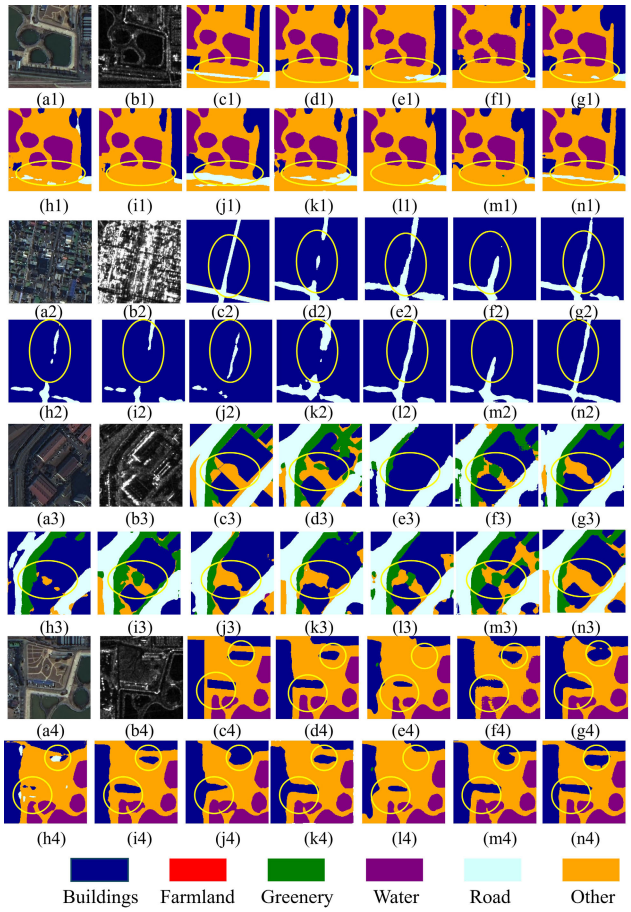


Fig. 7. Visualization of segmentation results on the Pohang dataset. (a1)–(a4) Optical images. (b1)–(b4) SAR images. (c1)–(c4) Ground truth. (d1)–(d4) SA-Gate. (e1)–(e4) V_FuseNet. (f1)–(f4) CFNet. (g1)–(g4) MCANet. (h1)–(h4) CMGFNet. (i1)–(i4) CMX. (j1)–(j4) AFNet. (k1)–(k4) CATNet. (l1)–(l4) GFBN. (m1)–(m4) FtransUnet. (n1)–(n4) Proposed method.

three datasets, with mIoU/OA improvements of 4.8%/2.3%, 2.6%/1.3%, and 2.4%/0.9% over the baseline. This validates the efficacy of our three proposed modules in facilitating deep cross-modal interactions to align multimodal features, mitigating intermodal disparities to effectively fuse complex multimodal information, and enhancing semantic segmentation through edge information utilization.

V. DISCUSSION

A. Overall Analyses of Datasets

To better explain the performance differences across the three datasets, we further analyze their characteristics and the segmentation challenges. The EG-CMFNet shows comparatively lower performance on the WHU-OPT-SAR dataset than the other two datasets, which is mainly attributed to two factors.

1) *Data Scale and Scene Diversity:* WHU-OPT-SAR contains 100 paired optical–SAR images, making its sample size tens of times larger than that of the other two datasets, exhibiting pronounced scene heterogeneity. This broad coverage leads to more complex land-cover distributions and higher interclass and intraclass variability, which increases the learning difficulty. Moreover, the dataset exhibits pronounced class imbalance, which imposes additional challenges during optimization.
2) *Spatial Resolution:* The WHU-OPT-SAR dataset has a spatial resolution of 5 m, resulting in more mixed pixels. Therefore, its interclass separability is weaker than that of the Pohang and Dongying datasets with 1-m spatial resolution. Despite these challenges, the proposed EG-CMFNet consistently outperforms comparative methods on all three datasets, indicating strong robustness under varying scales and scene complexities.

B. Effectiveness of MAFI at Different Stage

To further investigate the roles of each component in the MAFI module and the effect of sequential interaction, we conduct ablation experiments with a single interaction (channel-only or spatial-only) and with the dual-stage sequential design. The results show that the channel-only and spatial-only variants achieve similar performance, whereas the two-stage design yields better performance than the single-interaction setting. Moreover, channel-wise interaction preceding spatial interaction achieves the best segmentation results (Table V). We attribute this to channel interaction capturing global semantic dependencies while spatial interaction focuses on local structural features [71]. Since the combination of both dimensions enables the model to jointly capture global semantics and local information, the dual-stage interaction provides more discriminative feature representations than using either channel-only or spatial-only interaction. However, prioritizing local semantic feature extraction before global

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TABLE V
ABLATION STUDY FOR THE MAFI VARIANTS ON THE POHANG DATASET

MAFI Variants	mIoU(%)	OA(%)
Channel-only	94.2	97.3
Spatial-only	94.1	97.3
Spatial ⟶ Channel	94.6	97.5
Channel ⟶ Spatial	**95.0**	**97.7**

TABLE VI
ABLATION STUDY ON DIFFERENT INTERACTION MODES ON THE POHANG DATASET

Interaction Mode	mIoU(%)	OA(%)
without cross-modal interaction	94.6	97.5
MAFI(cross-modal interaction)	**95.0**	**97.7**

TABLE VII
ABLATION STUDY ON THE LEARNABLE PARAMETERS OF MAFI ON THE POHANG DATASET

	mIoU(%)	OA(%)
NO	94.2	97.3
L-G	94.6	97.5
Yes	**95.0**	**97.7**

Note: "NO": without learnable parameters; "L-G": linear gating in place of learnable parameters; "Yes": full setting with learnable parameters.

interaction risks submerging locally heterogeneous patterns under more homogeneous global contexts, thereby impeding the modeling of intricate spatial variations and compromising segmentation efficacy. In our method, multiscale pooled features are concatenated during the spatial interaction stage, which preserves global contextual information while reinforcing local details, ultimately leading to significant performance improvements.

C. Ablation Study on Different Interaction Modes

To further examine the role of cross-modal interactions, we compare MAFI with a variant without cross-modal interaction. In this variant, spatial and channel attention are computed independently within each modality. As shown in Table VI, introducing cross-modal interaction yields stable performance improvements. This may be because modeling attention independently within a single modality is relatively limited in suppressing modality heterogeneity and is less effective in alleviating local spatial misalignment between optical and SAR, thereby weakening intermodal consistency. In contrast, MAFI introduces cross-modal interaction in both the channel and spatial stages, allowing complementary information from the two modalities to mutually constrain each other and more effectively enhance consistent feature representations.

D. Ablation Study on the Learnable Parameters of MAFI

To further investigate the role of the learnable parameters in the MAFI module, we consider three comparative settings: removing the learnable parameters, replacing learnable parameters with a simple linear gating scheme, and using the full setting with learnable parameters. As shown in Table VII, removing these parameters leads to a noticeable performance drop, indicating that the model becomes less capable of reducing modality discrepancies. The linear gating scheme provides basic cross-modal fusion but lacks adaptive modulation of modal consistency, often suppressing information from one modality. In contrast, the model with learnable parameters achieves the best results, demonstrating that these parameters play a crucial role in reducing modality discrepancy and enhancing robustness.

TABLE VIII
ABLATION STUDY OF DIFFERENT MODALITIES ON THE POHANG DATASET

Mode	Data	mIoU(%)	OA(%)
Unimodal w/o EA	OPT	92.8	96.95
Unimodal w/o EA	SAR	86.5	94.34
Multimodal w/o EA	OPT+SAR	**94.1**	**97.39**

Note: w/o denotes without.

TABLE IX
ABLATION STUDY FOR THE FUSION MECHANISM ON THE POHANG DATASET

Method	mIoU(%)	Params(M)	FLOPs(G)	OA(%)
add	93.9	–	–	97.3
Concat	93.7	**6.26**	**1.208**	97.2
Crossattn	94.0	14.09	3.068	97.4
LFF	**95.0**	6.66	1.237	**97.7**

E. Ablation Experiments Between Different Modalities

We design a dual-stream architecture to fully extract and leverage complementary information across modalities, achieving more accurate segmentation. To evaluate this design and explore input modality impacts, ablation studies are conducted using optical-only inputs, SAR-only inputs, and optical–SAR fusion without edge branches (Table VIII). Results indicate superior performance with optical data versus SAR alone, as structural information in SAR data is more implicit and challenging to learn compared to textural or geometric features in optical images. Geometric distortion in SAR data further complicates spatial relationships. When fusing both modalities, our feature interaction modules effectively mitigate cross-modal discrepancies through deep exploitation of complementary high-level features, enabling multidimensional land-cover representation and enhanced segmentation accuracy. Quantitatively, fused data improves mIoU by 1.3% and OA by 0.44% versus optical-only inputs, validating that leveraging complementary modality information enables more effective segmentation.

F. Ablation Study With Fusion Mechanism

To explore the efficacy of different feature fusion mechanisms, we mark three baseline paradigms: feature summation, concatenation, and cross-attention fusion. All experiments maintain consistent input tensor dimensions. As shown in Table IX, the proposed LFF module achieves significant accuracy improvements, outperforming another method by

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TABLE X

ABLATION STUDY ON THE KERNEL SENSITIVITY OF LFF ON THE POHANG DATASET

	Kernels size	mIoU(%)	OA(%)
LFF_1	5	93.9	97.3
LFF_2	5,7	94.2	97.4
LFF_3	5,7,9	94.4	97.5
LFF_4 (Proposed)	5,7,9,11	**95.0**	**97.7**
LFF_5	5,7,9,11,13	93.9	97.2

TABLE XI

ABLATION STUDY ON THE SIMAM PLACEMENT SENSITIVITY OF LFF ON THE POHANG DATASET

SimAM Placement	mIoU(%)	OA(%)
Before fusion	94.3	97.4
After fusion	93.9	97.2
Before & After	**95.0**	**97.7**

Note: "Before fusion" and "After fusion" indicate applying SimAM only before or only after fusion, respectively; "Before & After" indicates applying SimAM both before and after fusion.

1.0%–1.3% in mIoU and 0.4% in OA on average. Parameter analysis indicates that the LFF module reduces parameters and computational complexity versus cross-attention while maintaining comparable costs to concatenation. These results demonstrate the superior capability of LFF module in suppressing noise and integrating complementary information, highlighting its architectural design for efficient fusion of multisource features.

G. Analysis of the Architectural Sensitivity of LFF

To further evaluate the structural sensitivity of the LFF module, we conduct additional ablation studies by varying the number and size of multiscale DW convolution kernels. As shown in Table X, when only a small number of DW convolution kernels are used, the relatively limited scale coverage may cause information loss during contextual understanding. Furthermore, we find that increasing the number of DW convolution kernel groups initially improves accuracy. However, beyond a certain point, introducing excessive kernel groups leads to a drop in performance. Among all the experiments, our four-group setting (5, 7, 9, and 11) is the most reliable. This setting improves mIoU by 0.6%–1.1%, indicating that introducing multiscale information in a reasonable manner can effectively promote segmentation performance. In addition, to examine the impact of SimAM placement on model performance, we conduct three comparative experiments: applying SimAM only before fusion, only after fusion, and both before and after fusion. As shown in Table XI, applying SimAM only before fusion yields limited gains, whereas applying it only after fusion performs worse due to insufficient suppression of early stage noise. In contrast, placing SimAM both before and after fusion achieves the best results, indicating that SimAM can effectively filter modality noise at both stages and thereby facilitate more reliable cross-modal fusion.

TABLE XII

COMPARATIVE EXPERIMENT ON THE EDGE BRANCH ON THE POHANG DATASET

Edge Branches	mIoU(%)	OA(%)	Params (M)
SAR	94.2	97.4	**3.55**
OPT	**95.0**	**97.7**	3.56
SAR+OPT	94.4	97.5	6.45

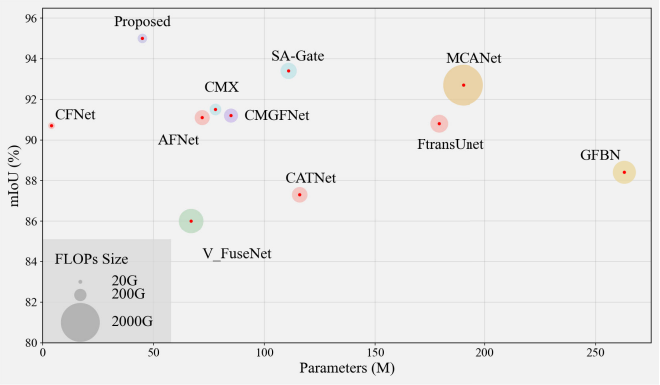


Fig. 8. Model parameters and computational complexity.

H. Comparative Experiment on Different Edge Branches

To investigate the effectiveness of edge feature extraction from different image sources, we conduct a comparative experiment on the imagery type processed by the edge branch. As shown in Table XII, the results indicate that extracting edge features from the optical branch contributes more substantially to semantic segmentation performance than extracting them from the SAR branch. This is mainly because optical images usually provide clearer edge details, whereas boundaries in SAR images often are discontinuous due to speckle noise. In addition, using edge branches for both optical and SAR does not bring further improvement compared with using optical edges only while increasing the parameter count. This may be attributed to modality discrepancies that lead to inconsistent boundary predictions between the two modalities, thereby introducing interference during edge guidance.

I. Model Complexity Analysis

Model accuracy and efficiency are critical for evaluating overall model performance. Table XIII and Fig. 8 present computational efficiency versus accuracy on the Pohang dataset for all semantic segmentation methods. It can be seen that our method achieves the best segmentation performance while maintaining low parameters and FLOPs. Specifically, regarding parameter size, our proposed method is only larger than CFNet, and concerning FLOPs, it is slightly higher than CATNet and CMX. However, it delivers significantly improved accuracy over these methods. Compared to the majority of competing methods, our approach achieves comprehensive superiority in both computational efficiency and segmentation accuracy. Concretely, the parameters of our model represent only 25.2% of those required by FtransUnet, 17% of GFBN, and 41% of SA-Gate, among others. Overall,

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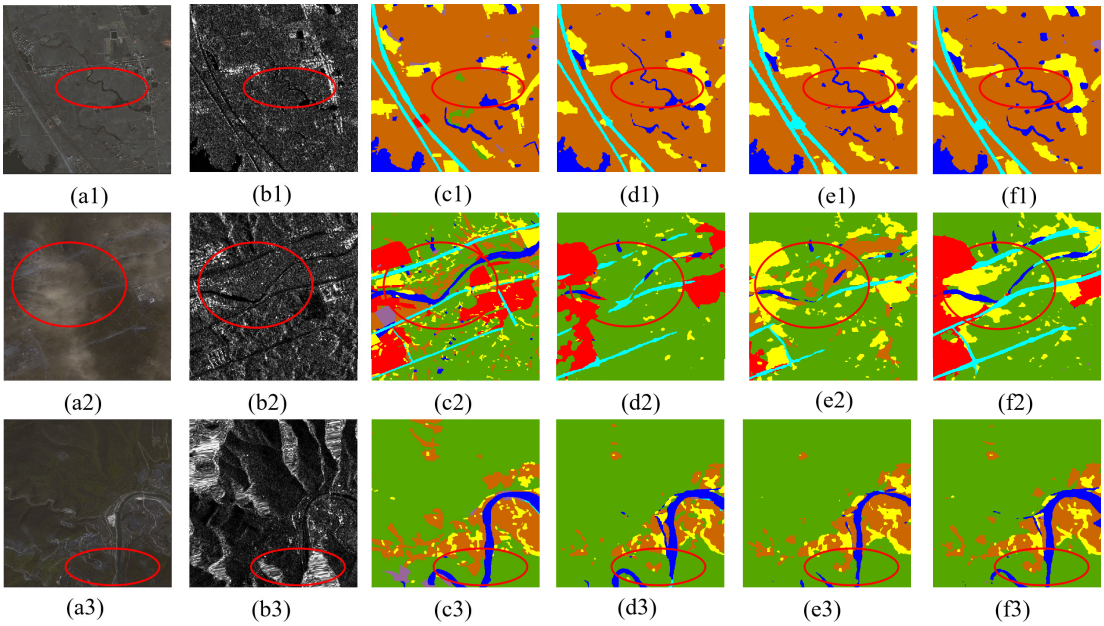


Fig. 9. Restricted scenarios for the segmentation of the proposed model. (a1)–(a3) Optical images. (b1)–(b3) SAR images. (c1)–(c3) Ground truth. (d1)–(d3) AFNet. (e1)–(e3) CMX. (f1)–(f3) Proposed method.

TABLE XIII
CALCULATION EFFICIENCY RESULTS ON THE POHANG DATASET

Method	Params (M) ↓	FLOPs (G) ↓
SA-Gate	110.9	330.1
V_FuseNet	66.7	692.2
CFNet	**4.3**	**78.9**
MCANet	189.9	3035
CMGFNet	85.2	312.6
CMX	77.4	128.8
AFNet	71.3	260.8
CATNet	115.7	143.8
GFBN	262.7	615.5
FtransUnet	179.4	451.1
Proposed	45.3	237.8

Note: ↓ denotes lower values are better.

we effectively balance computational burden while attaining the best segmentation results. Given the feature extraction capability and strong segmentation performance, we conclude that EG-CMFNet holds significant advantages for multisource remote sensing image semantic segmentation applications.

J. Limitations and Future Work

Although our framework achieves the best performance in multimodal remote sensing images, its performance may degrade under several challenging scenarios.

1) *Annotation Noise and Label Errors:* In some regions, the annotations are inconsistent with the visual appearance. As highlighted in Fig. 9(a), mislabeled objects may negatively affect the evaluation of EG-CMFNet predictions. In some cases, these predictions are more consistent with the actual surface morphology than the provided labels, which may lead to an underestimation of the true performance of the proposed model.
2) *SAR-Dominant Scenarios:* As shown in Fig. 9(b), the optical image is occluded by clouds, while the corresponding SAR image contains extensive weak-scattering areas. When the model lacks sufficiently discriminative information, the effectiveness of cross-modal feature interaction decreases, which weakens multimodal fusion and may lead to missed or incomplete segmentation.
3) *Severe Cross-Modal Spatial Inconsistency:* As illustrated in Fig. 9(c), optical images can clearly present the distribution of water bodies in a region, whereas the corresponding SAR images exhibit inconsistent characteristics due to side-looking imaging and terrain-induced effects. This cross-modal inconsistency directly degrades the segmentation performance. Despite these limitations, EG-CMFNet remains more robust than comparative methods in terms of overall structure preservation and object boundary integrity in these challenging cases.

To further overcome the above challenges, we will improve our method in two aspects in the future. First, since multimodal data can effectively alleviate missing modality information, we will extend our framework to incorporate more modalities, such as LiDAR and DEM data, to further validate the effectiveness of our proposed method. In addition, introducing more modalities may increase spatial misalignment across modalities; therefore, we will explore combining explicit and implicit alignment strategies to mitigate spatial biases in multimodal data and further improve cross-modal alignment.

VI. CONCLUSION

In this article, we propose a multistream semantic segmentation framework that integrates CNN and Transformer architectures to address the challenges of complementary information fusion and boundary blurring in multimodal remote sensing imagery. Experiments on three benchmark

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datasets of different scales demonstrate that our method achieves state-of-the-art performance and strong generalization ability. These results verify the effectiveness of the proposed components. The MAFI module employs a channel–spatial adaptive interaction mechanism to better exploit complementary multimodal cues and alleviate modality heterogeneity and spatial misalignment. The LFF module captures both global and local features, adapting to diverse object shapes and scales with high computational efficiency. The EA module enhances boundary discrimination by reducing confusion between adjacent objects and preserving structural continuity. Although further research is still needed to more fully exploit the potential of multimodal data and improve model interpretability, the proposed framework lays a solid foundation for future work. As the volume of remote sensing satellite data continues to grow, deep learning approaches based on multimodal inputs are becoming increasingly important. Our developed model offers a novel perspective on multimodal semantic segmentation, particularly in the fusion of optical and SAR data, and demonstrates strong potential to advance the development of the multimodal remote sensing paradigm.

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REFERENCES

[1] Q. Zhu et al., "Land-use/land-cover change detection based on a Siamese global learning framework for high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 63–78, Feb. 2022.

[2] K. Ni, Y. Xie, G. Zhao, Z. Zheng, P. Wang, and T. Lu, "Coarse-to-fine high-order network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5509716.

[3] P. Zhang et al., "A hybrid attention-aware fusion network (HAFNet) for building extraction from high-resolution imagery and LiDAR data," *Remote Sens.*, vol. 12, no. 22, p. 3764, Nov. 2020.

[4] H. Luo, C. Chen, L. Fang, K. Khoshelham, and G. Shen, "MS-RRFSegNet: Multiscale regional relation feature segmentation network for semantic segmentation of urban scene point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8301–8315, Dec. 2020.

[5] Z. He, P. Wang, B. Huang, D. Zhu, H. F. Lee, and H. Leung, "Dadigan: A dual attention blocks-based disentangled iterative generative adversarial network for cloud and shadow removal on SAR and optical images," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103487.

[6] G. Sun, J. Cheng, A. Zhang, X. Jia, Y. Yao, and Z. Jiao, "Hierarchical fusion of optical and dual-polarized SAR on impervious surface mapping at city scale," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 264–278, Feb. 2022.

[7] Y. Zhang, H. Zhang, and H. Lin, "Improving the impervious surface estimation with combined use of optical and SAR remote sensing images," *Remote Sens. Environ.*, vol. 141, pp. 155–167, Feb. 2014.

[8] J. Han et al., "Prediction of winter wheat yield based on multi-source data and machine learning in China," *Remote Sens.*, vol. 12, no. 2, p. 236, Jan. 2020.

[9] G. Sun, X. Rong, A. Zhang, H. Huang, J. Rong, and X. Zhang, "Multi-scale Mahalanobis kernel-based support vector machine for classification of high-resolution remote sensing images," *Cognit. Comput.*, vol. 13, no. 4, pp. 787–794, Jul. 2021.

[10] X. Gao and S. Liu, "BCMFIFuse: A bilateral cross-modal feature interaction-based network for infrared and visible image fusion," *Remote Sens.*, vol. 16, no. 17, p. 3136, Aug. 2024.

[11] F. Lang, M. Zhang, J. Zhao, N. Zheng, and H. Shi, "Semantic segmentation for multisource remote sensing images incorporating feature slice reconstruction and attention upsampling," *Int. J. Remote Sens.*, vol. 45, no. 8, pp. 2761–2785, Apr. 2024.

[12] C. Hazirbas, L. Ma, C. Domokos, and D. Cremers, "FuseNet: Incorporating depth into semantic segmentation via fusion-based CNN architecture," in *Proc. Asian Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 213–228.

[13] B. Ren, B. Liu, B. Hou, Z. Wang, C. Yang, and L. Jiao, "SwinTFNet: Dual-stream transformer with cross attention fusion for land cover classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[14] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[15] Q. Zhu et al., "Oil spill contextual and boundary-supervised detection network based on marine SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5213910.

[16] A. Xu, Z. Xue, Z. Li, S. Cheng, H. Su, and J. Xia, "UM2Former: U-shaped multimixed transformer network for large-scale hyperspectral image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5506221.

[17] C. Peng, X. Zhang, G. Yu, G. Luo, and J. Sun, "Large kernel matters—Improve semantic segmentation by global convolutional network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4353–4361.

[18] Y. Wang, Y. Li, G. Wang, and X. Liu, "Multi-scale attention network for single image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 5950–5960.

[19] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[20] W. Wang, X. Tan, P. Zhang, and X. Wang, "A CBAM based multi-scale transformer fusion approach for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6817–6825, 2022.

[21] C. Liu et al., "A review of optical and SAR image deep feature fusion in semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 12910–12930, 2024.

[22] K. Li, Q. Geng, M. Wan, X. Cao, and Z. Zhou, "Context and spatial feature calibration for real-time semantic segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 5465–5477, 2023.

[23] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet-A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[24] X. Chen et al., "Bi-directional cross-modality feature propagation with separation-and-aggregation gate for RGB-D semantic segmentation," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*. Glasgow, U.K.: Springer, Aug. 2020, pp. 561–577.

[25] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 14679–14694, Dec. 2023.

[26] H. Ma, H. Yang, and D. Huang, "Boundary guided context aggregation for semantic segmentation," 2021, *arXiv:2110.14587*.

[27] X. Zheng, L. Huan, G.-S. Xia, and J. Gong, "Parsing very high resolution urban scene images by learning deep ConvNets with edge-aware loss," *ISPRS J. Photogramm. Remote Sens.*, vol. 170, pp. 15–28, Dec. 2020.

[28] D. Marmanis, K. Schindler, J. D. Wegner, S. Galliani, M. Datcu, and U. Stilla, "Classification with an edge: Improving semantic image segmentation with boundary detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 158–172, Jan. 2018.

[29] S. S. Shivakumar, N. Rodrigues, A. Zhou, I. D. Miller, V. Kumar, and C. J. Taylor, "PST900: RGB-thermal calibration, dataset and segmentation network," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 9441–9447.

[30] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021.

[31] H. Hosseinpour, F. Samadzadegan, and F. D. Javan, "CMGFNet: A deep cross-modal gated fusion network for building extraction from very high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 96–115, Feb. 2022.

[32] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

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[33] X. Ma, X. Zhang, and M.-O. Pun, "A crossmodal multiscale fusion network for semantic segmentation of remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3463–3474, 2022.
[34] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.
[35] S. Cui, W. Chen, W. Xiong, X. Xu, X. Shi, and C. Li, "SiMultiF: A remote sensing multimodal semantic segmentation network with adaptive allocation of modal weights for Siamese structures in multiscene," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4406817.
[36] X. Li et al., "MCANet: A joint semantic segmentation framework of optical and SAR images for land use classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102638.
[37] Y. Niu, L. Tan, L. Zhang, and X. Zuo, "TMFNet: Two-stream multi-channels fusion networks for color image operation chain detection," 2024, *arXiv:2409.07701*.
[38] C.-C. Zhang and J.-D. Fang, "Edge detection based on improved Sobel operator," in *Proc. Int. Conf. Comput. Eng. Inf. Syst.*, 2016, pp. 129–132.
[39] X. Wang, "Laplacian operator-based edge detectors," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 5, pp. 886–890, May 2007.
[40] J. Canny, "A computational approach to edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vols. PAMI-8, no. 6, pp. 679–698, Nov. 1986.
[41] R. Chen, F.-L. Zhang, and T. Rhee, "Edge-aware convolution for RGB-D image segmentation," in *Proc. 35th Int. Conf. Image Vis. Comput. New Zealand (IVCNZ)*, Nov. 2020, pp. 1–6.
[42] S. Gao, X. Yang, L. Jiang, Z. Fu, and J. Du, "Global feature-based multimodal semantic segmentation," *Pattern Recognit.*, vol. 151, Jul. 2024, Art. no. 110340.
[43] B. Sui, Y. Cao, X. Bai, S. Zhang, and R. Wu, "BIBED-seg: Block-in-block edge detection network for guiding semantic segmentation task of high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1531–1549, 2023.
[44] W. Zhou, S. Dong, C. Xu, and Y. Qian, "Edge-aware guidance fusion network for RGB–thermal scene parsing," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 3, pp. 3571–3579.
[45] S. Dong, W. Zhou, C. Xu, and W. Yan, "EGFNet: edge-aware guidance fusion network for RGB–thermal urban scene parsing," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 657–669, Jan. 2024.
[46] L. Zhu, X. Wang, Z. Ke, W. Zhang, and R. W. Lau, "BiFormer: Vision transformer with bi-level routing attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10323–10333.
[47] P. Zhang, B. Peng, C. Lu, Q. Huang, and D. Liu, "ASANet: Asymmetric semantic aligning network for RGB and SAR image land cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 574–587, Dec. 2024.
[48] Z. Du, X. Li, J. Miao, Y. Huang, H. Shen, and L. Zhang, "Concatenated deep-learning framework for multitask change detection of optical and SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 719–731, 2024.
[49] W. Wu, S. Guo, Z. Shao, and D. Li, "CroFuseNet: A semantic segmentation network for urban impervious surface extraction based on cross fusion of optical and SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2573–2588, 2023.
[50] S. Xiao, P. Wang, W. Diao, K. Fu, and X. Sun, "A multimodal semantic segmentation framework for heterogeneous optical and complex SAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 8083–8098, 2025.
[51] N. Audebert, B. Le Saux, and S. Lefèvre, "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 140, pp. 20–32, Jun. 2018.
[52] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[53] T. Zhao et al., "Artificial intelligence for geoscience: Progress, challenges, and perspectives," *Innov.*, vol. 5, no. 5, 2024, Art. no. 100691.
[54] F. Zhao, C. Zhang, and B. Geng, "Deep multimodal data fusion," *ACM Comput. surveys*, vol. 56, no. 9, pp. 1–36, 2024.
[55] Y. Xiao, T. Xu, X. Yu, Y. Fang, and J. Li, "A lightweight fusion strategy with enhanced interlayer feature correlation for small object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4708011.
[56] D. Ye, J. Peng, W. Guo, and H. Li, "Adaptive multiscale slimming network learning for remote sensing image feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5647513.
[57] L. Yang, R. Zhang, L. Li, and X. Xie, "SimAM: A simple, parameter-free attention module for convolutional neural networks," in *Proc. 38th Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 11863–11874.
[58] J. Li, J. Zhu, C. Li, X. Chen, and B. Yang, "CGTF: Convolution-guided transformer for infrared and visible image fusion," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–14, 2022.
[59] J. Yang, P. Qiu, Y. Zhang, D. S. Marcus, and A. Sotiras, "D-Net: Dynamic large kernel with dynamic feature fusion for volumetric medical image segmentation," 2024, *arXiv:2403.10674*.
[60] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, "Gated-SCNN: Gated shape CNNs for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5229–5238.
[61] X. Li, L. Xie, C. Wang, J. Miao, H. Shen, and L. Zhang, "Boundary-enhanced dual-stream network for semantic segmentation of high-resolution remote sensing images," *GIScience Remote Sens.*, vol. 61, no. 1, Dec. 2024, Art. no. 2356355.
[62] L. Huang, M. Wang, B.-H. Tang, W. Le, and Q. Tian, "CIBENet: A channel interaction and bridging-enhanced change detection network for optical and SAR remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 131, Jul. 2024, Art. no. 103969.
[63] Y. Wang et al., "Extracting building footprints in SAR images via distilling boundary information from optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5204215.
[64] L. Wan, H. Zhang, G. Lin, and H. Lin, "A small-patched convolutional neural network for mangrove mapping at species level using high-resolution remote-sensing image," *Ann. GIS*, vol. 25, no. 1, pp. 45–55, Jan. 2019.
[65] D. Ouyang et al., "Efficient multi-scale attention module with cross-spatial learning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.
[66] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.
[67] B. Ren et al., "A dual-stream high resolution network: Deep fusion of GF-2 and GF-3 data for land cover classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102896.
[68] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[69] W. Kang, Y. Xiang, F. Wang, and H. You, "CFNet: A cross fusion network for joint land cover classification using optical and SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1562–1574, 2022.
[70] F. Sun, P. Ren, B. Yin, F. Wang, and H. Li, "CATNet: A cascaded and aggregated transformer network for RGB-D salient object detection," *IEEE Trans. Multimedia*, vol. 26, pp. 2249–2262, 2024.
[71] Z. Chen, Y. Zhang, J. Gu, L. Kong, X. Yang, and F. Yu, "Dual aggregation transformer for image super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 12312–12321.



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